Index

IE Autospriter Interface....2
Closed form....2
Step 1: Global Parameters....2
Step 2: QoL Option – Quality of Life(CHANGED)....3
Step 3: Camera....5
Step 4: Animation....6
Collections....7
Step 5: Final....8
Requirements....8
Helpful Insights....9
Save file Cycles 4.5.5 LTS....14
Scene Collections....14
Render Engine....14
Camera settings....15
Sun....15
Plane....16
Area....16
Save file EEVEE 4.5.5 LTS....17
Render Engine....17
Plane....17
Shading Materials....18
Demo files 4.5.5(TODO)....18
Animation Type 0000....19
Note....19
Animation Type 2000(CHANGED)....19
Note....20
Animation Type 4000....21
Animation Type B000....21
Animation Type 7000 Monster....24
No Weapon Overlays....24
Notes....24
Weapon Overlays....25
Shading Nodes....27
Notes....27
Animation Type 7000 Monster_old....28
Animation Type 8000(NEW)....29
Notes....29
Animation Type 9000(NEW)....30
Notes....30
Animation Type C000....30
Blender Settings....31
Shading Nodes....32
Notes....32
Animation Type D000....34
Troubleshooting/FAQ....34
„Save at“ path empty, what happens?....34

How can time performance be increased?....34
Usage of GPU power....34
Alter sampling parameters....35
Use Decimate Modifier....35
Change Render Engine....36
Specific animation types spam new images in Image Editor....37
How does the resolution setting work?....37
Don’t use special characters for the string properties....37
Blender terminated unexpectedly during rendering....38

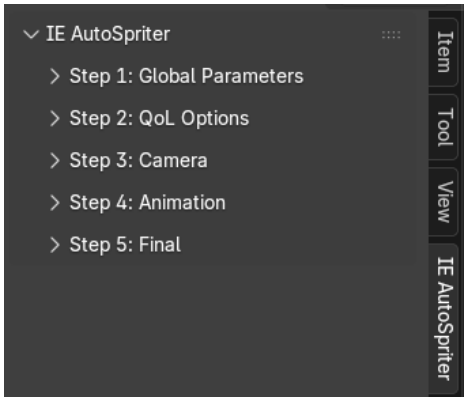
Note: The content of this manual may change and therefore may not be applicable to every version of IE AutoSpriter. It also serves as a preliminary manual. I'm also not a professional Blender user or Blender add-on developer. I learn by doing.

ALWAYS make a backup file of your project before rendering with IE AutoSpriter.

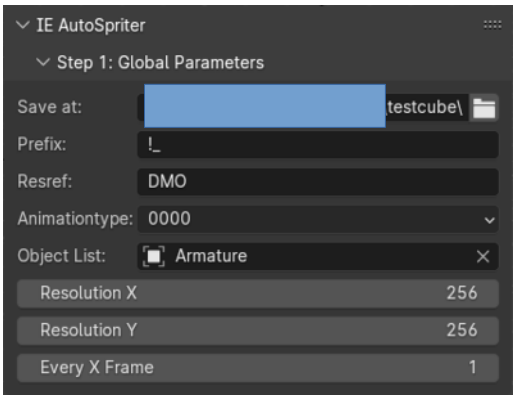
IE Autospriter Interface

This Blender add-on automates the process of rendering sprites specifically for Infinity Engine animations. The workflow is divided into logical steps to guide the user, though the sequence of the first four steps is flexible. Step 5 initiates the rendering process. Using or knowing the iesdp documentation is essential to understanding how and why things are designed the way they are.

Closed form



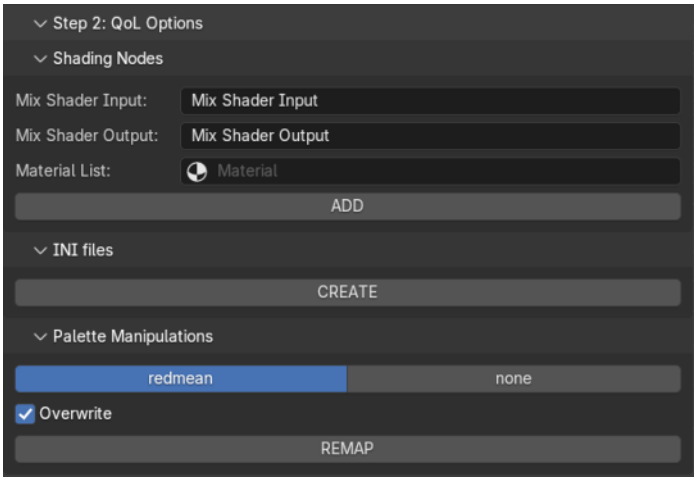
Step 1: Global Parameters



This step defines general settings that apply across the entire sprite rendering process.

- **Save at:** Defines the directory where the generated sprite files will be saved.
- **Prefix:** A user-definable modder prefix that will be added to the beginning of the sprite file names.
- **Resref (Resource Reference):** This is the freely definable part of the sprite file name. Other parts of the file names are fixed and automatically generated by the add-on.
- **Object List:** This must be the armature that contains the NLA tracks. The NLA tracks contain all the animations for the armature (essentially, your creature's animation).
- **Animationtype:** is a dropdown menu that lets you choose the **type of creature or object** you're rendering based on iesdp. Its purpose is to dynamically show or hide the correct set of options in the subsequent steps (*`Step 3: Camera`* and *`Step 4: Animation`*), ensuring that the UI only presents relevant choices for the selected animation type.
- **Resolution X:** This refers to the image width resolution
- **Resolution Y:** This refers to the image height resolution
- **Every X frame:** Saves every xth frame as a sprite. This can be useful, for example, when there are too many frames to save as sprites. The steps depend on the frame start.

Step 2: QoL Option – Quality of Life(CHANGED)



This optional step allows for the addition of specific shaders to the material shading setup. This is crucial for Infinity Engine sprites, which require indexed color palettes.

Mix Shader Input/Output: These string property fields are used to specify the names of existing shader nodes within the material. The new, automatically added shader nodes will be placed and connected between these specified nodes. The added shader nodes are „Brightness/Contrast“ and „Mix Shader“. This prevents pure black(rgb 0,0,0) texture color.

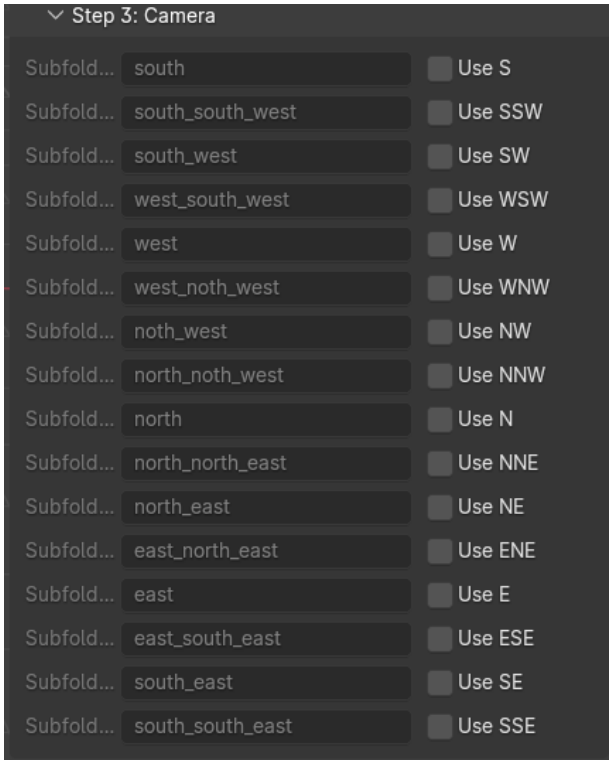
CREATE button: This **button** creates a predefined and finished **INI** file. It is 'predefined' because the rendering or the **files do not need to exist yet**, and 'finished' because it pulls information directly from the **Blender** project and **IE AutoSpriter** to generate the file. It requires the correct information from steps 1, 3, and 4. The animations, meshes, and collections **must already be selected (via the checkboxes)**.

redmean: Uses the redmean formula to calculate weighted color distances in the RGB color space, better matching human visual perception.

Overwrite: Determines whether existing output files are directly overwritten during the remapping process or saved as new files.

REMAP: Triggers the palette remapping process and processes the images based on the selected settings (metric and overwrite mode). Note that the location of the rendered sprites is set by parameters in steps 1, 3, and 4. These parameters include "Save at" and activated "Subfolder" checkboxes. The feature prechecks whether the target folders exist to prevent errors during the remapping process. Remapping is a post-rendering process, meaning the sprite images must already be rendered before this feature can be used.

Step 3: Camera

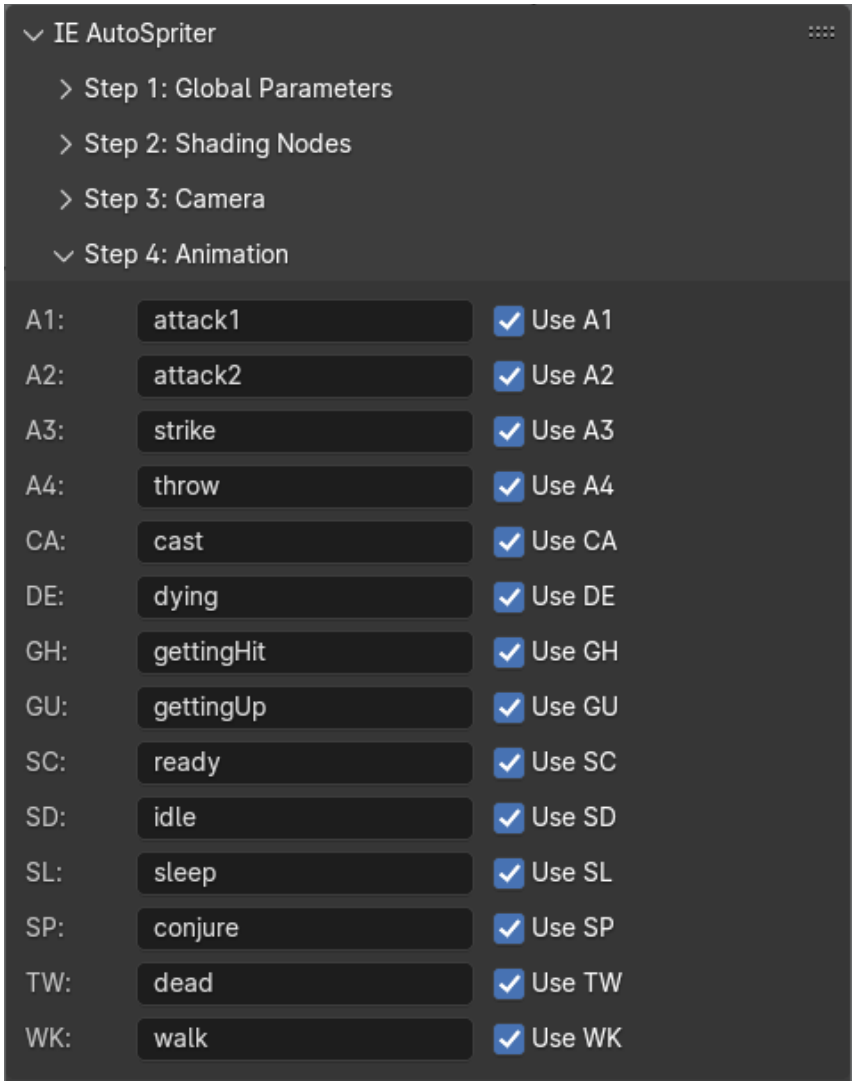




This step manages the output folders and defines which camera orientations will be rendered.

- Orientation Folders: These checkboxes determine whether sprites for a specific orientation will be rendered and saved into their corresponding designated folders. The available orientations are **dynamically displayed based on sthe `Animationtype` selected in Step 1**. If a checkbox is not activated, sprites for that particular orientation will be ignored and not rendered.
- Rotation: The numbers indicate the degree of rotation, and the curved arrows show clockwise or counterclockwise rotation. Use the "**ROTATE (Preview)**" button to start the rotation. The rotation can be applied during the running object animation. This serves as a preview; the rotation parameters are not applied during rendering.

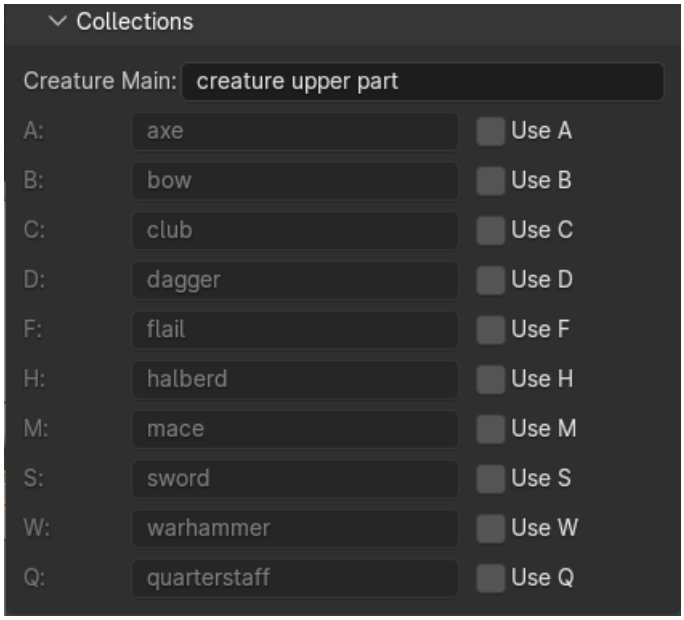
Step 4: Animation

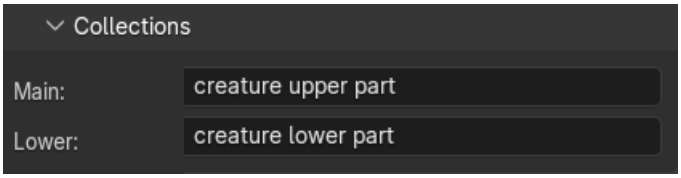


This step defines which animations (Blender Actions) should be rendered and how they are named in the output.
The available animations/actions are **dynamically displayed based on the `Animationtype` selected in Step 1**. If a checkbox is not activated, sprites for that particular animation/action will be ignored and not rendered.

- **Animation(Blender Action):** This refers to the type of animation, corresponding to an "Action" in Blender (e.g., "A1" for Attack Animation 1).
- **Checkbox:** If activated, the specified animation type (e.g., "A1") will be appended to the sprite file name.
- **String Input Field:** The name entered in this input field must precisely match the name of the corresponding Action in Blender.

Collections



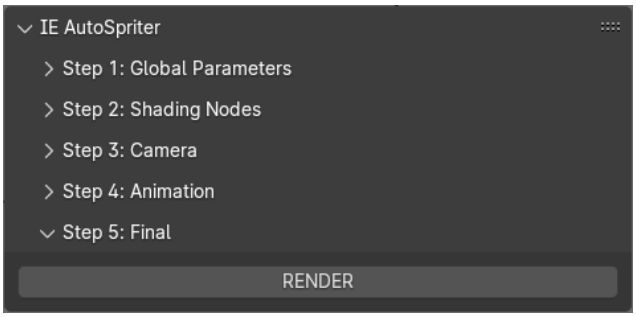


The available weapon animations are **dynamically displayed based on the `Animationtype` selected in Step 1**. If a checkbox is not activated, sprites for that particular orientation will be ignored and not rendered.

- **Main:** Specifies the name of the Blender collection containing the main creature model or upper part of effect model(type 0000). This is an absolutely necessary input!
- **Lower:** Specifies the name of the Blender collection containing the creature's lower model or lower part of effect model(type 0000). This is an absolutely necessary input for animation type 3000! For Animation Type 0000, leaving this field blank triggers a toggle like effect that prevents the rendering of the lower section. In addition to the collection name, it also serves as the folder name for the lower section of the export pipeline.
- **A, B, C, etc. (String inputs)**: Define the names of the Blender collections for each weapon type (e.g., "axe", "bow", "club"). The key names(A,B,C, etc.) are also used as wovl (weapon overlay) identifiers in filenames.
- **Use A, Use B, Use C, etc.**: Enable or disable rendering of sprites for each specific weapon collection.

The path for a weapon sprite is as follows(windows): <**Save at**>\<**Weapon Collection**>\ <**Subfolder position**>

Step 5: Final

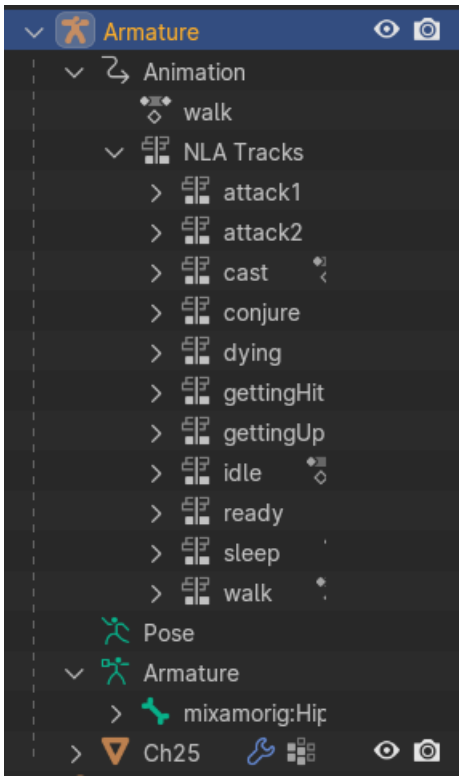


This is the concluding step that triggers the rendering process.

Render: Upon activation, this step initiates the rendering and saving of all sprites according to the settings configured in the preceding steps.

Requirements

It is necessary that the animation is listed in the **Armature** within the **Animation** object. This means that under **Animation** the actions are collected in the **NLA Tracks** (see example in the image below).

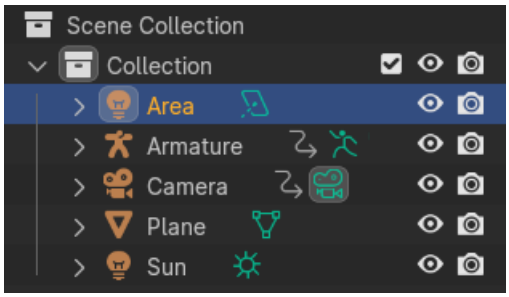


The reason for this is that **IE AutoSpriter uses this list to sequentially specify actions (type of animation) to render**.

The collection containing the main armature for the creature must be activated.

In addition, the names of the animation folders in **step 4** must exactly match the names found in the animation object (see **NLA Tracks**).

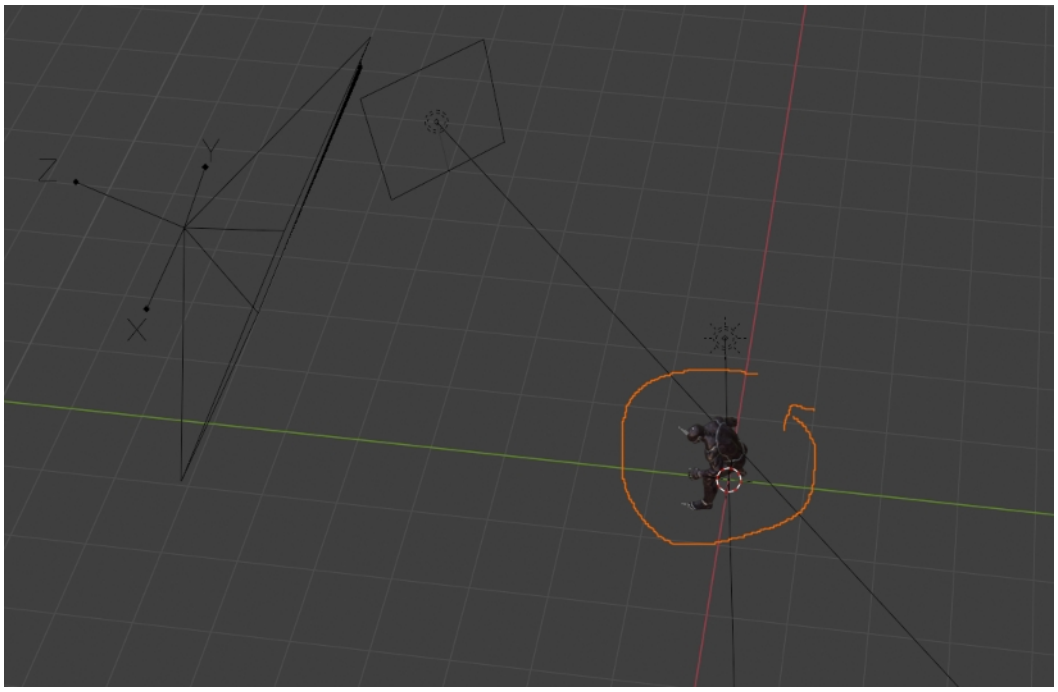
The blend file needs the following setup:



- **Camera**: This is needed to render the creature model.
- **Armature**: This has already been discussed.
- **Plane**: This is needed to create a shadow for the creature model.
- **Area**: This is a light source that should be adjusted as desired.
- **Sun**: This is needed to create a shadow on the **Plane**.

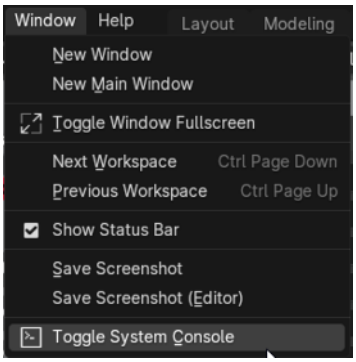
Helpful Insights

The different angles of the creature model are not created by moving the camera, but by rotating the model while the camera remains in a fixed position.



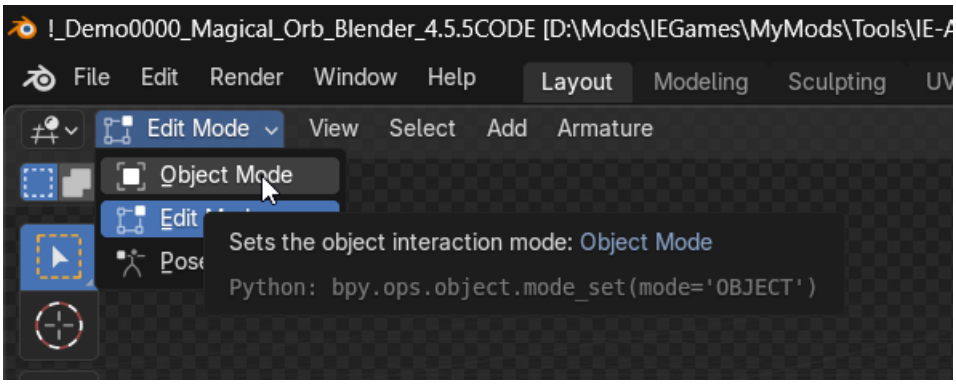
The objects in the collection have a number of specific important parameters that are not covered here, as the details can be found in the save files (these may be covered in later versions of IE AutoSpriter).

To view the rendering progress and elapsed time, go to "**Toggle System Console**" (see image below). This should be done **BEFORE** rendering, as rendering makes it impossible to interact with Blender.



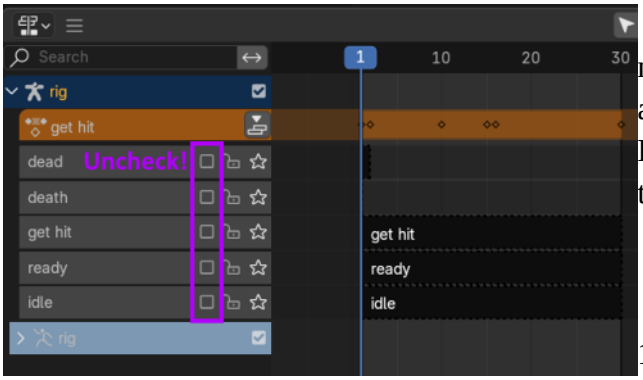
After rendering, the creature's position is reset to the position before rendering started.

Object Mode:

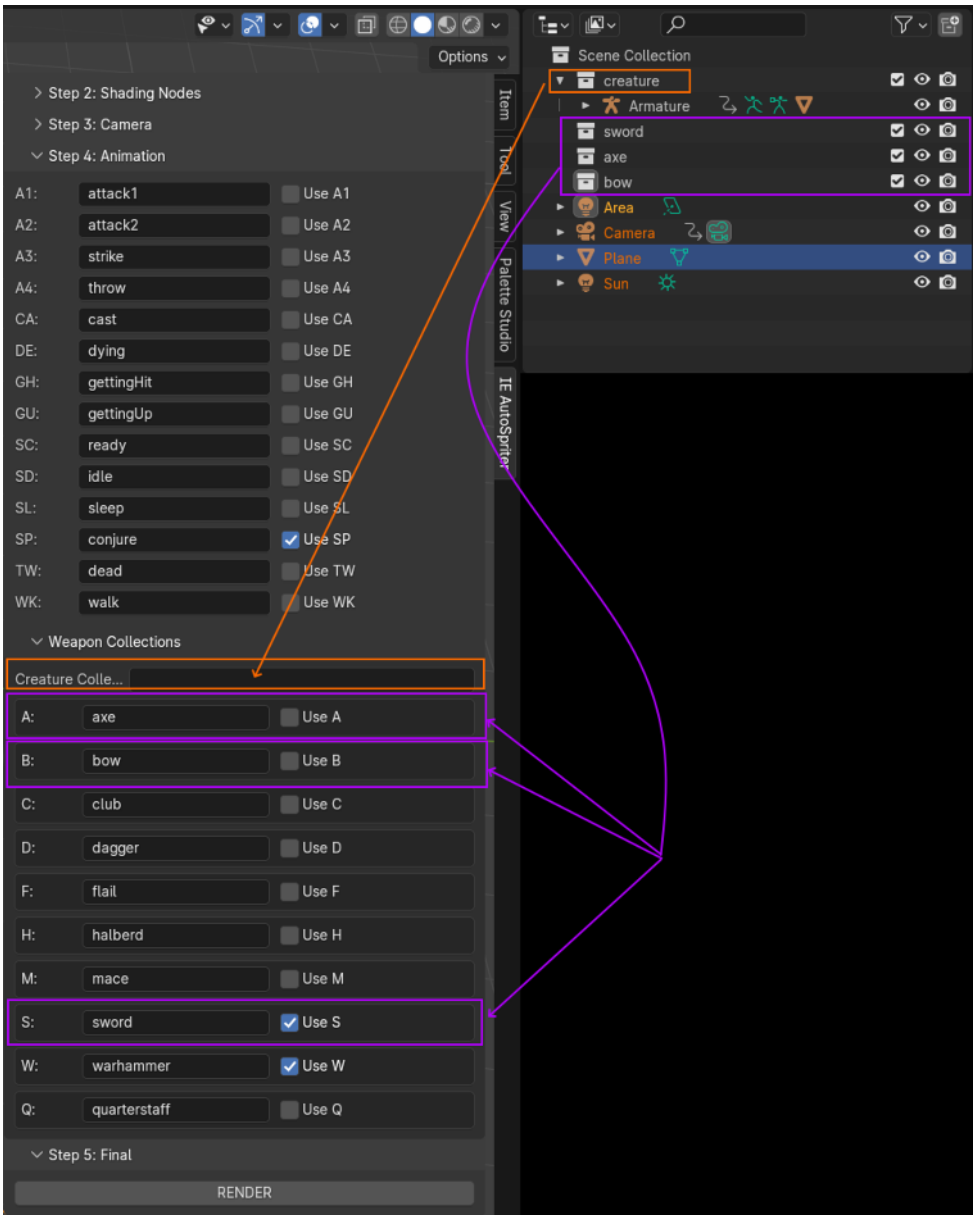


The "3D viewport" must be in "object mode", otherwise a corresponding error message will be displayed when the "RENDER" button is pressed.

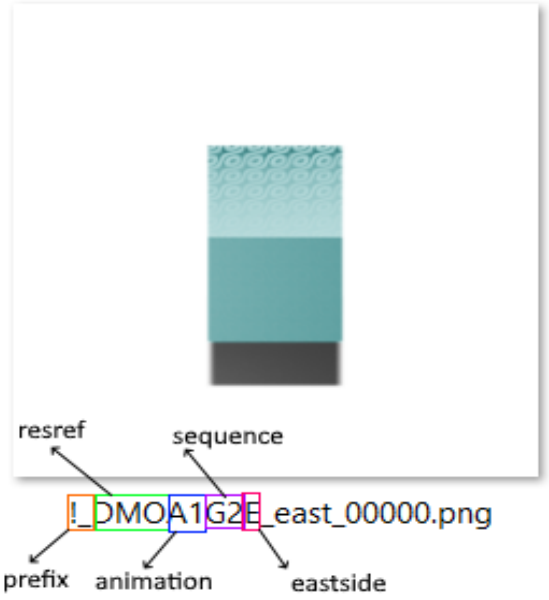
Nonlinear Animation:



To play all the animations of the NLA tracks, they must first be muted. Then, the desired animation/action can be selected via the "Action Editor," and all animations will play without mixed together.



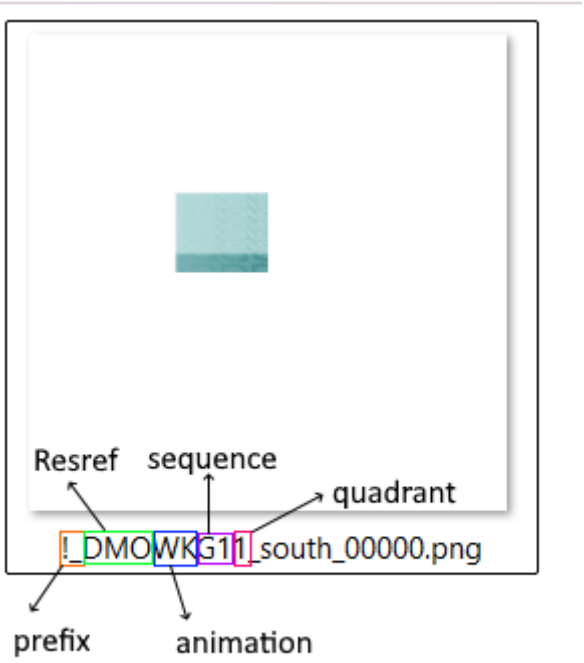
Each weapon to be rendered requires its own collection. This is (at the moment) only required for type E000. Everything else, such as area, camera, plane, and sun, can or should be outside of other subcollections. Depending on the lighting (sun), the weapon rendering may contain shadows of both the weapon and the creature. This shouldn't be a problem, since the creature and weapon shadows are merged when the BAM files are combined anyway, and since the weapon and creature shadows (must) be the same black color, the difference shouldn't be visible. This is only theoretical, as no tests have been conducted so far.



Character Naming

The add-on uses a class called IEAS_AnimationTypes to determine which characters to use for each animation. These characters (e.g., G1, G2, G3, and E) serve to identify the specific sequence a file belongs to. The E character, specifically, is used for "Eastside animations" and is added to the filename depending on the Animationtype. The user doesn't need to manually input these characters; the add-on handles this internally.

For animation types that handle quadrants, the handling is slightly different. The quadrant number is added it can be identified which quadrant it belongs to. For the 1000 monster quadrant type, the quadrants are divided into 4 equal parts with the resolution of the original image.



The filename construction depends on the animation type, as some do not have sequences or quadrants.

In general, if an animation type requires sprite splitting (image/animation splitting), the rendering resolution must be divisible by 2 or 3. 1000 monster quadrants require a resolution divisible by 2, and 1000 monster multi must be divisible by 2 for "split_bams=0" and by 3 for "split_bams=1."

Animation Type 5000/6000 character old

This type is special in that its sequences (regarding file naming) are inconsistent. It is strongly recommended that you consult the iesdp documentation to understand the exact reasons for this.

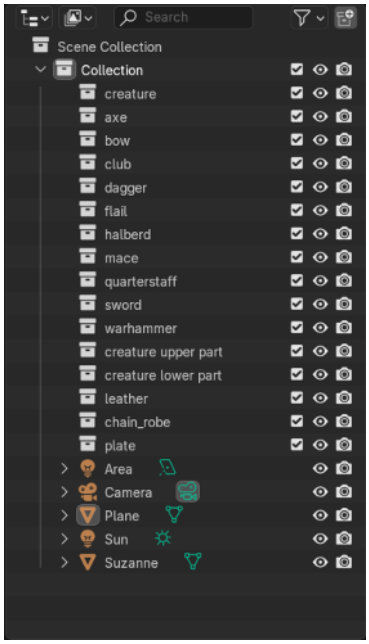
Two points are particularly important: First, this animation type has the W2 and W2E sequences but does not utilize the available orientations, while the other sequences do not use the W2 and W2E orientations. This is relevant because, although IE AutoSpriters offers all 16 orientations when this type is selected, it uses them internally according to the iesdp documentation.

For example, no Walk2 animation sprites are generated for the "South" orientation.

Save file Cycles 4.5.5 LTS

Found in „**IE-AutoSpriter-\save files\!_Template_CYCLES_Blender_4.0.blend**“

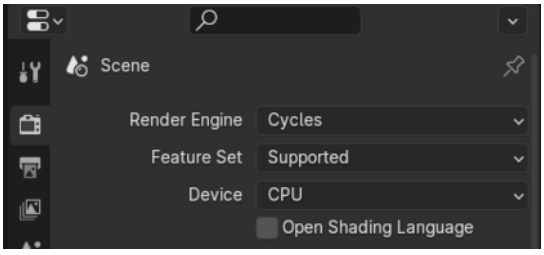
Scene Collections



„creature“ is generally the main collection, which should be used for creature animations. The creature collection should also be used if no weapon animations are specified. Empty scenes should not cause rendering issues.

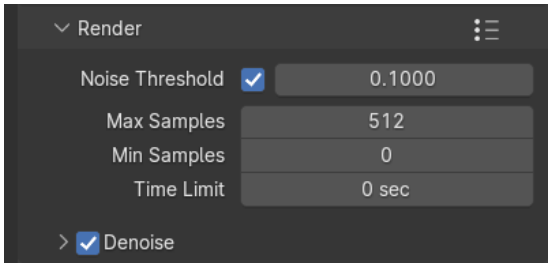
For animation type 3000, the "**creature upper part**" and "**creature lower part**" collections are relevant. The others can be deleted.

Render Engine

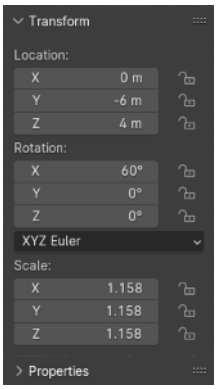


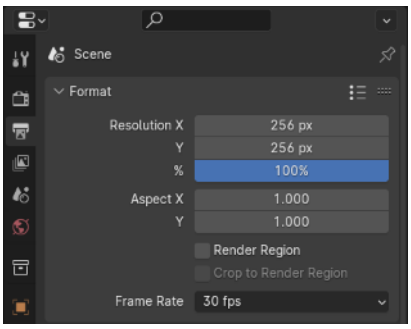
The rendering engine is set to Cycles (instead of EEVEE). This allows the "**Plane**" object to be rendered transparently without the "**Sun**" object losing its ability to cast shadows.

The render settings for the engine is as follows(see next image below):



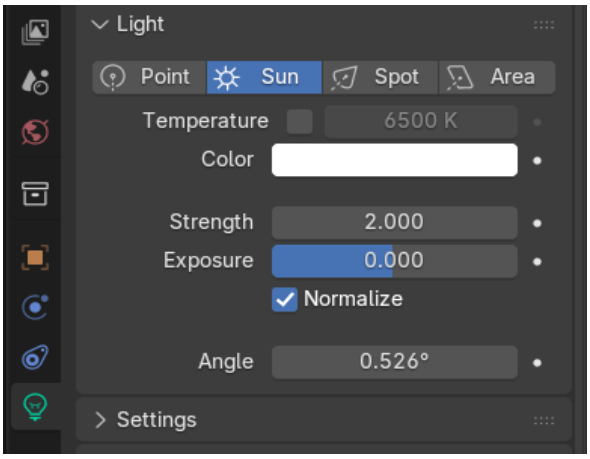
Camera settings

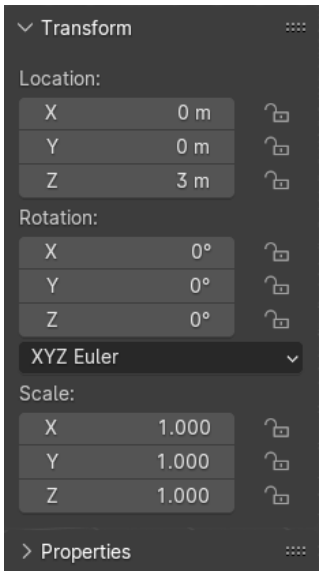




The camera settings can be adjusted to suit your personal preferences if needed. For example, the camera angle is set to 60°, although I've heard arguments for 45°. The rendering resolution is set to **256px to 256px**(px stands for pixels) and frame rate to **30fps**(frames per second).

Sun

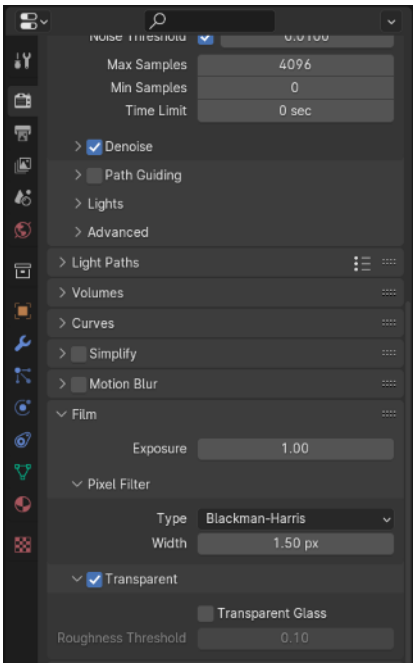


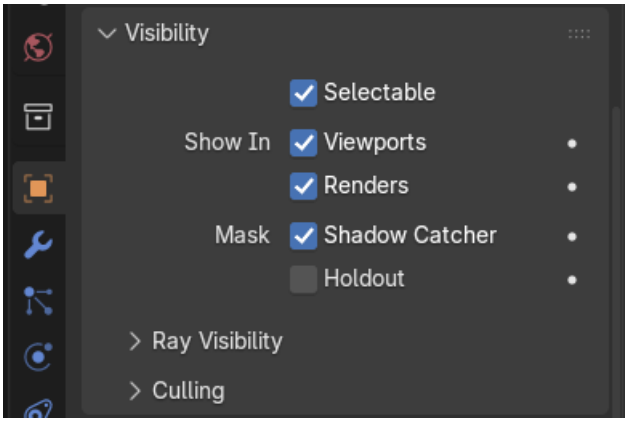


These settings are optional but worked reasonably well (**lumens 2.000 and Z-distance 3m**), but will likely need to be adjusted depending on the model that needs to be rendered.

Plane

The object „**Plane**“ is used to capture the shadow cast by sunlight on an object. To prevent the plane from being included in rendering without losing the shadow, the "**Transparent**" switch is enabled under „**Rend → Film**“ checked(see left screenshot). To use the pane as shadow catcher, it needs to be enabled as such(see right screenshot).





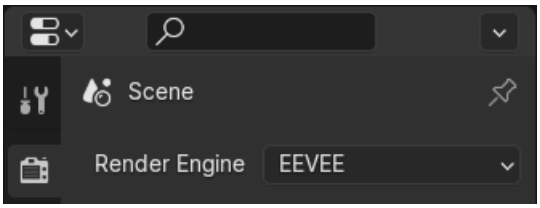
Area

The „**Area**“ object is a purely optional light source. Its main purpose is to make the model's texture more visible.

Save file EEVEE 4.5.5 LTS

Found in „**IE-AutoSpriter-\save files\!_Template_EEVEE_Blender_4.5.5.blend**“ and is similar to cycles save file.

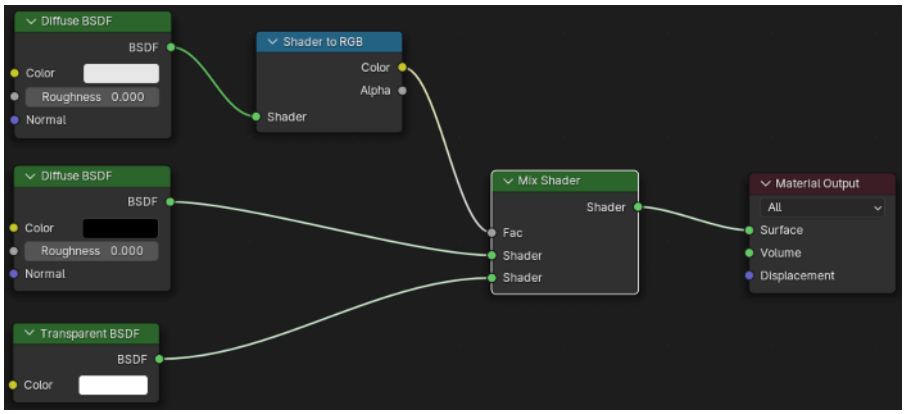
Render Engine



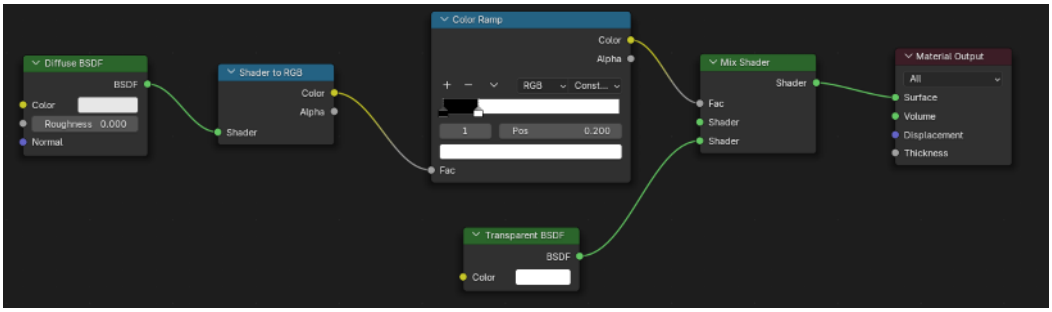
The rendering engine is set to Cycles (instead of EEVEE). This significantly reduces rendering time(made with blender 4.0) compared to Cycles. However, it also brings its own challenges for the „**Plane**“.

Plane

The object „**Plane**“ is used to capture the shadow cast by sunlight on an object. To make everything work, the layer has its own materials and shading nodes within them (see images below).



Option 1: *Shadow_Catcher2_Material*



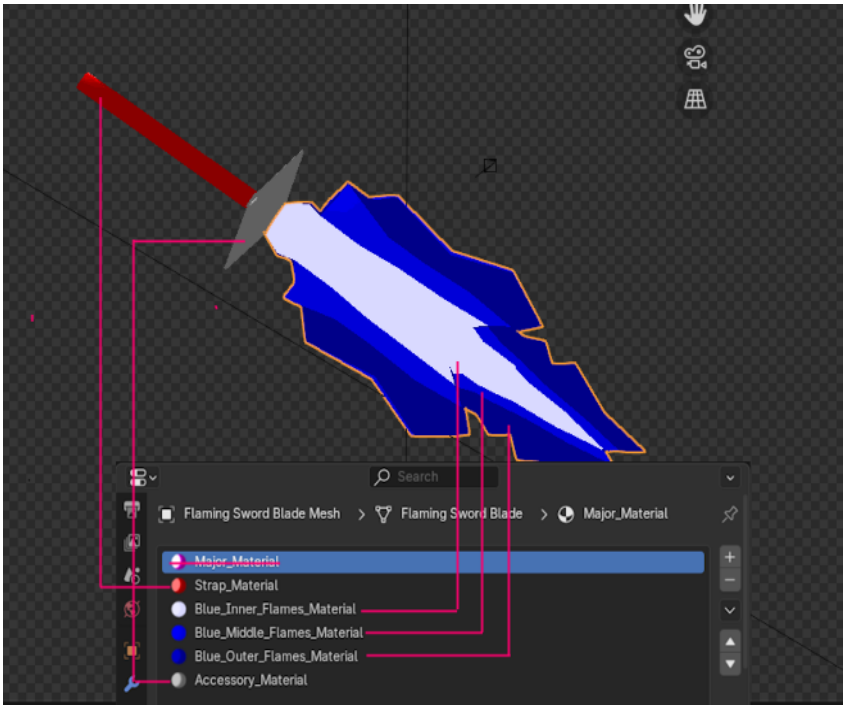
Option 2: *Shadow_Catcher1_Material*

This was realized with the help of the following video:

- How to make a shadow catcher in Blender 4.0 Cycles and Eevee by MK Graphics.
- Blender 4.4.0 Shadow Catcher Tutorial – Eevee

Shading Materials

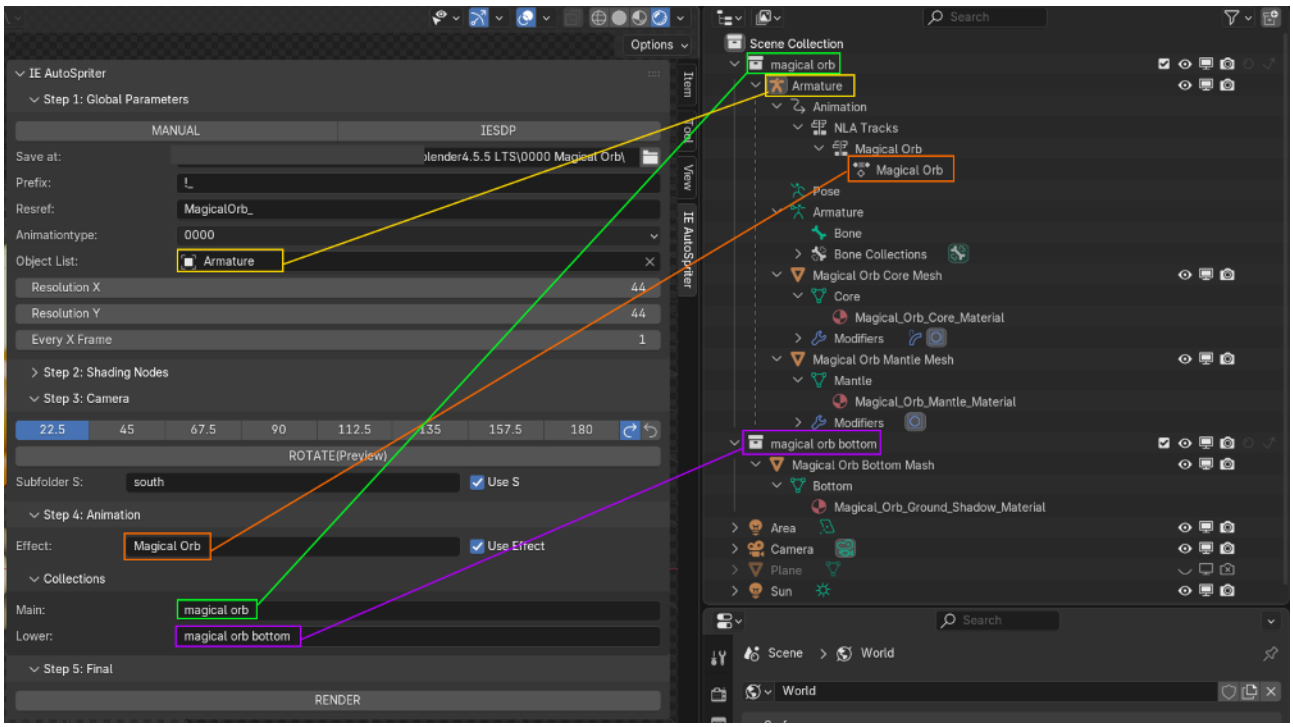
The save contains various shading materials regarding the color replacements. Here follows an exmaple(‚*!_Demo7000_Fishman_Blender_4.5.5.blend*‘ - flaming sword):



Demo files 4.5.5(TODO)

They are intended primarily to demonstrate/test the features of IE Autospriter.

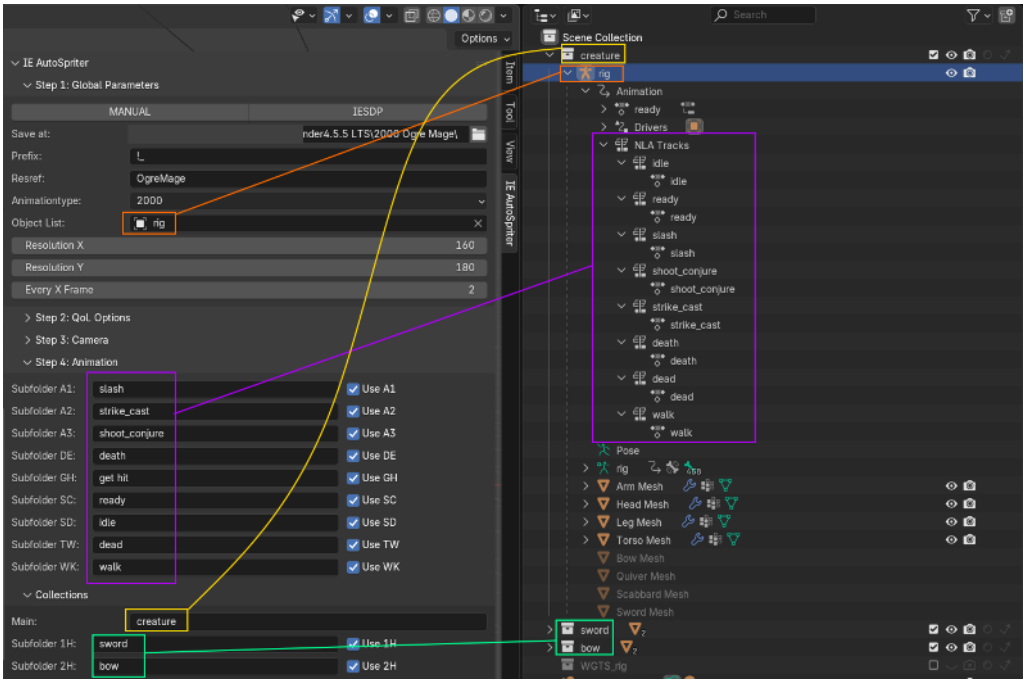
Animation Type 0000



Note

This needs to be tested more thoroughly, but it is recommended to place both collections directly under "Scene Collection", generally for animation type 0000.

Animation Type 2000(CHANGED)



Note

The blend file parameters are set as shown here in „Animation Type C000“.

Furthermore there some issue regarding iesdp informations. It seems there are not entirely correct.

Details can be found here: https://www.gibberlings3.net/forums/topic/41396-test-results-for-animation-type-2000-monster-layered-spell/#comment-360188

1.) It appears that SD and SC are swapped in this position. The correction in iesdp would be as follows:

G1: WK=0-4, SD=8-12, SC=16-20, GH=24-28, DE=32-36, TW=40-44

G1E: WK=5-7, SD=13-15, SC=21-23, GH=29-31, DE=37-39, TW=45-47

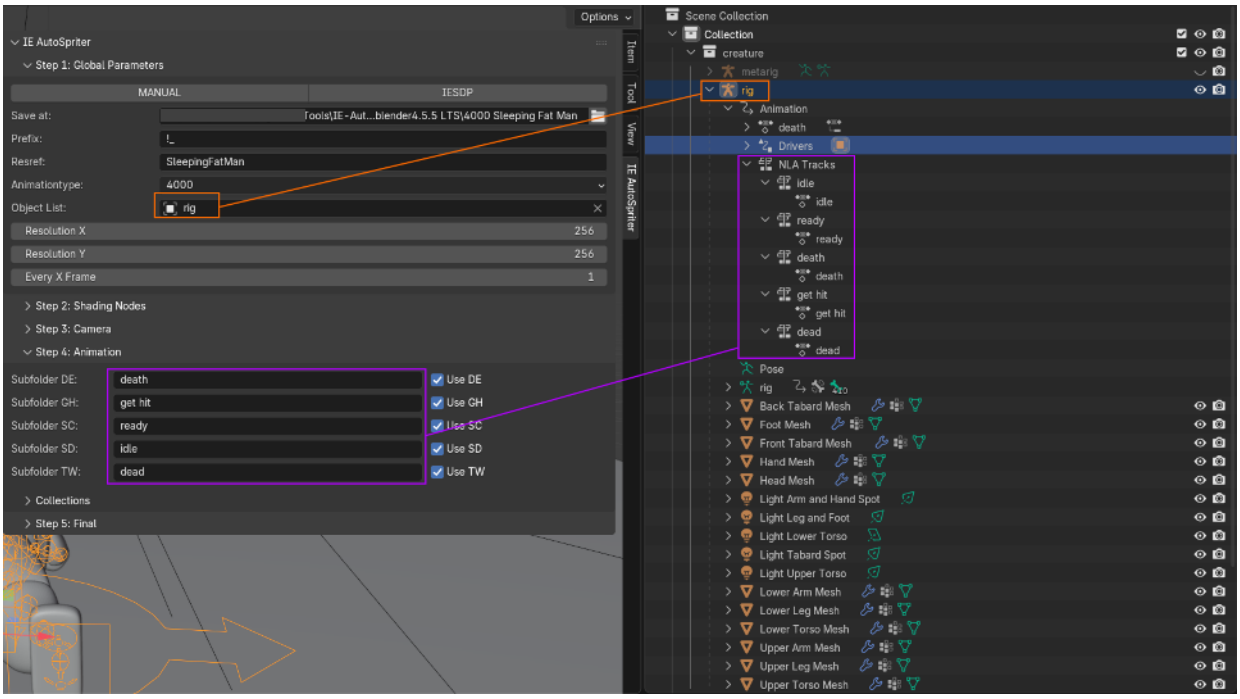
2.) Furthermore A2/CA=8-12 is played correctly when spell is cast, but while casting the idle animation is played, (but as idle and ready animation are swapped, it is the ready animation that is played). Either way expected behavior is playing A3/SP=16-20

3.) It seems, the weapons were layered differently in terms of their orientation for longbows it seems. Behaviour is triggered by the item "Left-handed" flag.

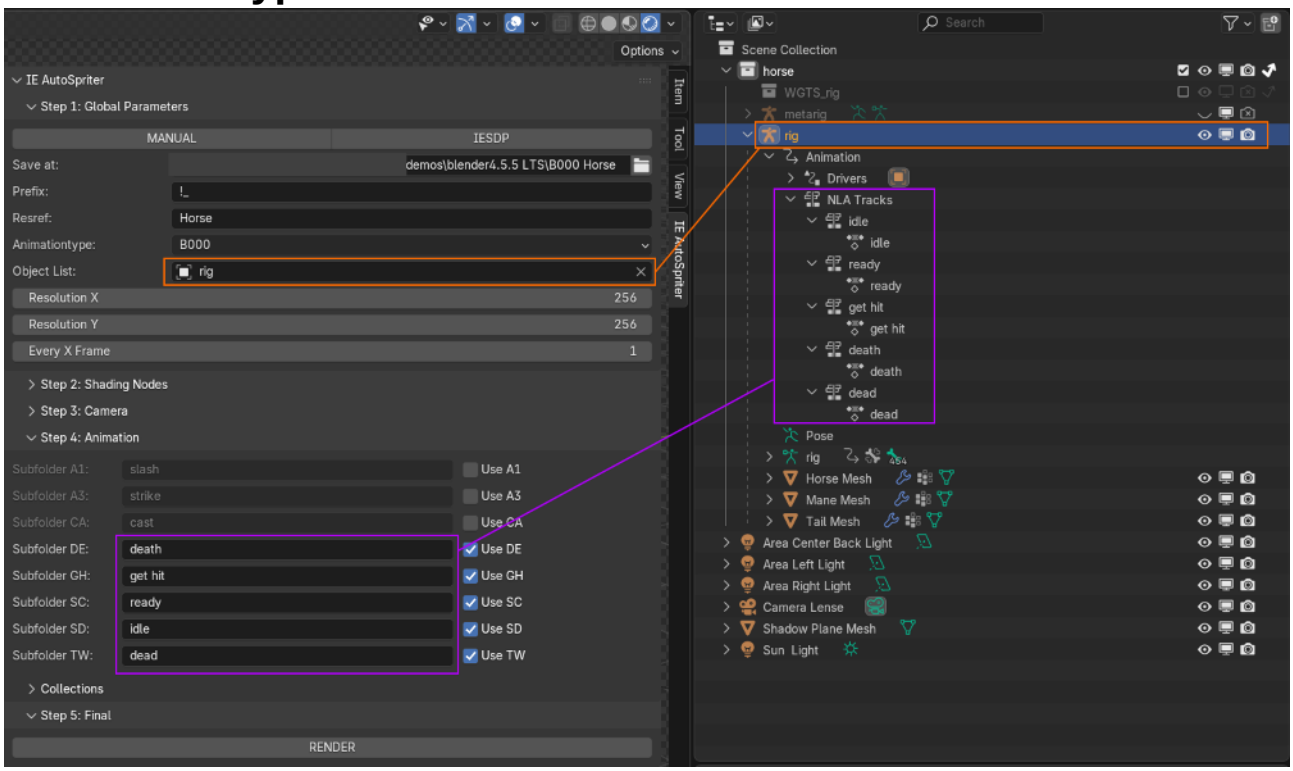
- 0 → S: WP=Front, CH=Back,
- 1 → SW: WP=Front, CH=Back,
- 2 → W: WP=Front, CH=Back,
- 3 → NW: WP=Front, CH=Back,
- 4 → N: WP=Back, CH=Front,
- 5 → NE: WP=Back, CH=Front,
- 6 → E: WP=Back, CH=Front,
- 7 → SE: WP=Back, CH=Front,

Animation Type 4000

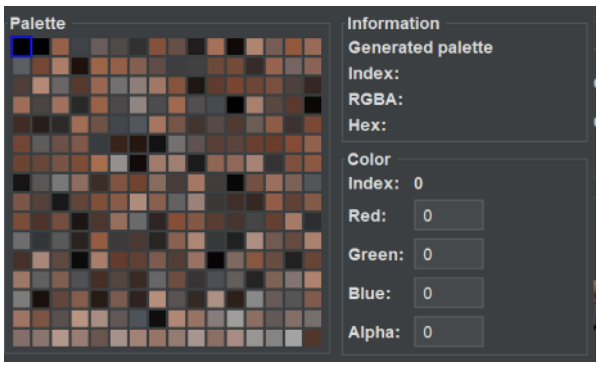
The Blender settings are the same as for "Animation Type C000". The same applies to the shading nodes, except that more materials were used here.

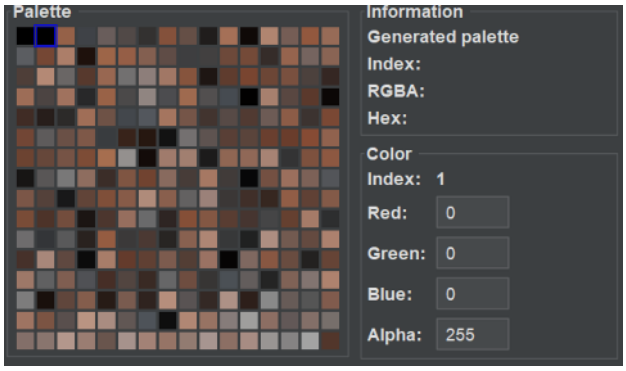


Animation Type B000



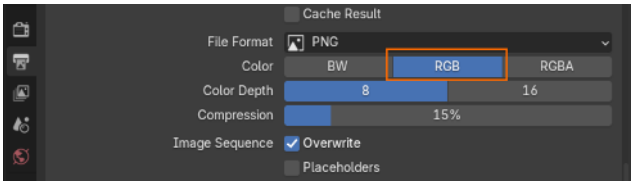
~~Screenshots of Near Infinity:~~





~~This is not the actual color palette of the BAM files, but the logic still seems to hold true.~~

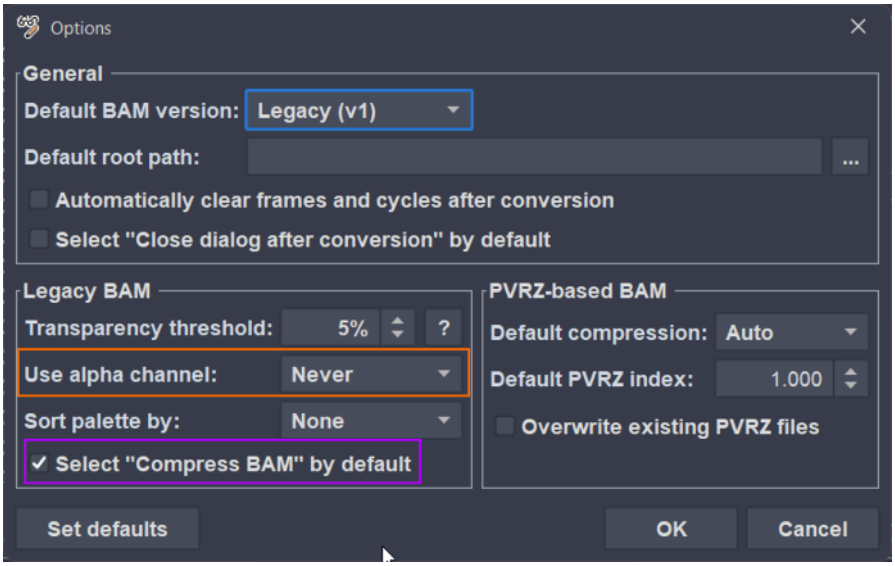
~~The settings for index 0 and index 1 are relevant for the color palette. The color palette can be achieved by rendering a single image, such as a "dead" animation (containing only a single frame), in **RGB**. This image is loaded into the Near Infinities BAM Converter, and the palette needs then to be locked or saved as a seperate BAM file which then can be exported for its color palette.~~

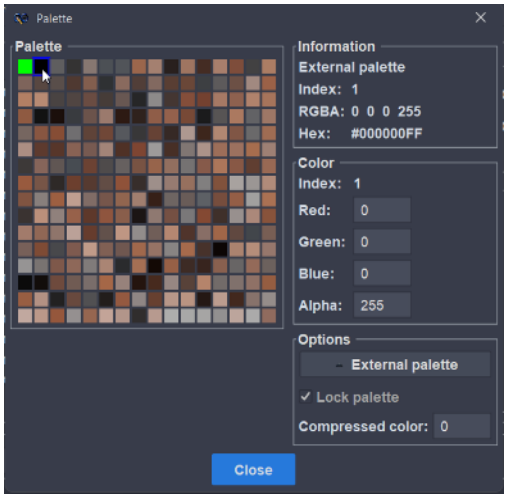


~~Next, remove this image and add all frames created with **RGBA** to the still-open BAM converter. Then unlock the palette, change the color indices 0 and 1 as shown in the screenshots, and lock it again.~~

~~This has been successfully tested twice, but requires further investigation/testing.~~

Although the stroke process described above works mostly, the following process in NI is easier to follow.

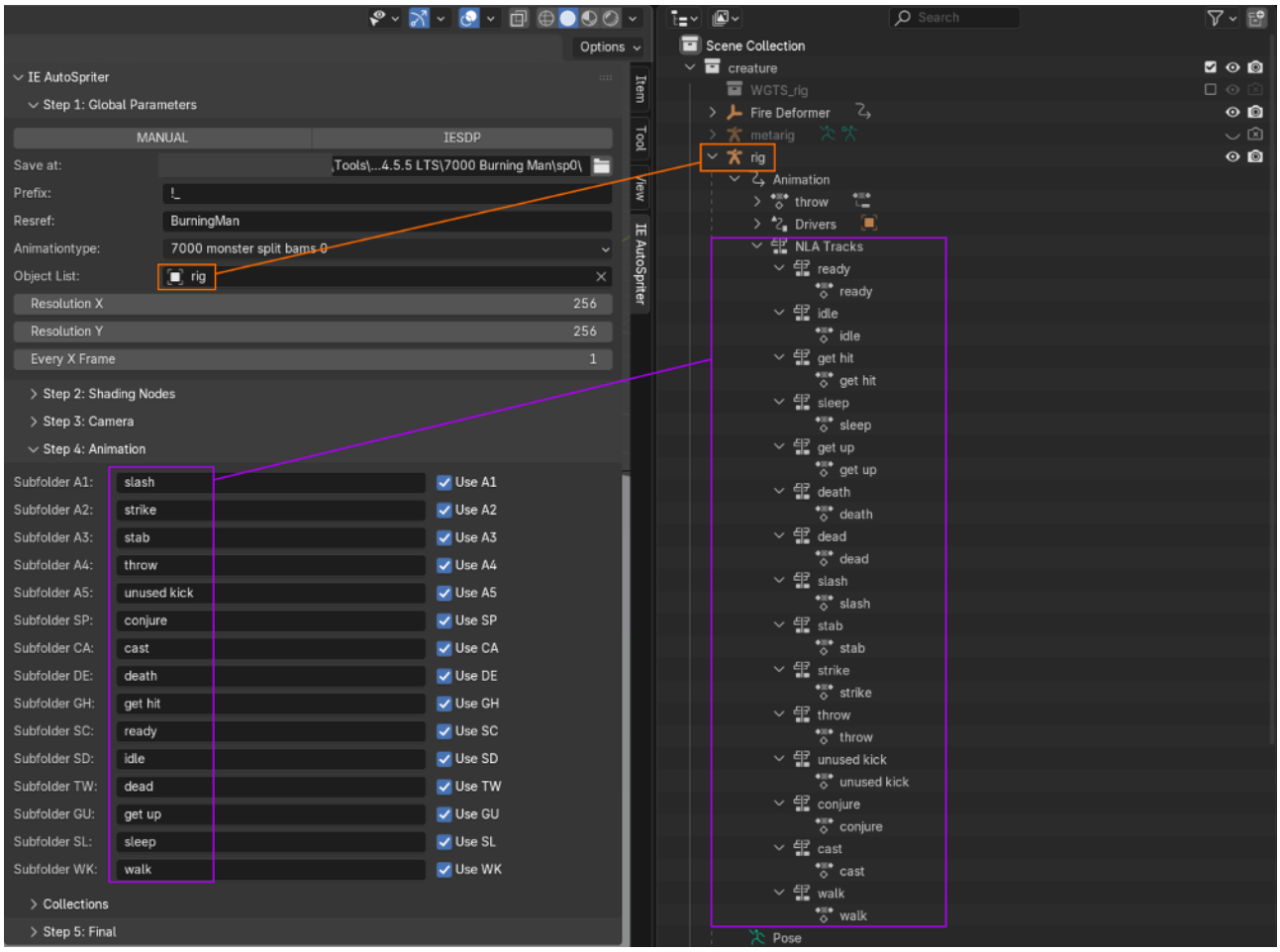




Selecting 'Never' for the alpha channel produced a color palette with a value of 255. In this demo, it fit well enough for the sprite representation within the game. Checking 'Compress BAM' reduced the size of the BAMs significantly. Also, color index 1 needs to be black for the shadow of the creature to be transparent, as the shadow is black.

Animation Type 7000 Monster

No Weapon Overlays



Notes

See for more detailed information: https://www.gibberlings3.net/forums/topic/41144-test-results-for-animation-type-7000-monster/

Split_bams=0:

- A4=27-35, A5=36-44 Both work, but depend on being triggered by a ranged weapon. There is one exception for the A4 and A5 formats: Dart. Darts will play A4 by default, but will change to A5 when a sling is used. This persists until another weapon using A4 resets the dart animation.
- SL=54-62, GU=63-71 Despite testing, these animations could not be triggered.

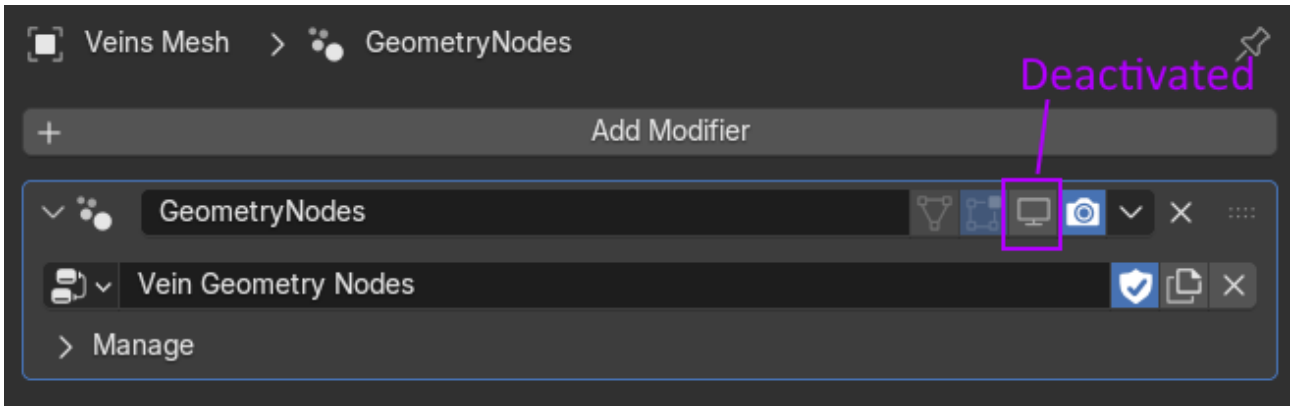
Split_bams=1:

- G24: A5=36-44 is not used because it is not triggered by the ranged weapons tested (bow, throwing dart, slingshot, crossbow, throwing dagger and throwing axe). Instead, the attack

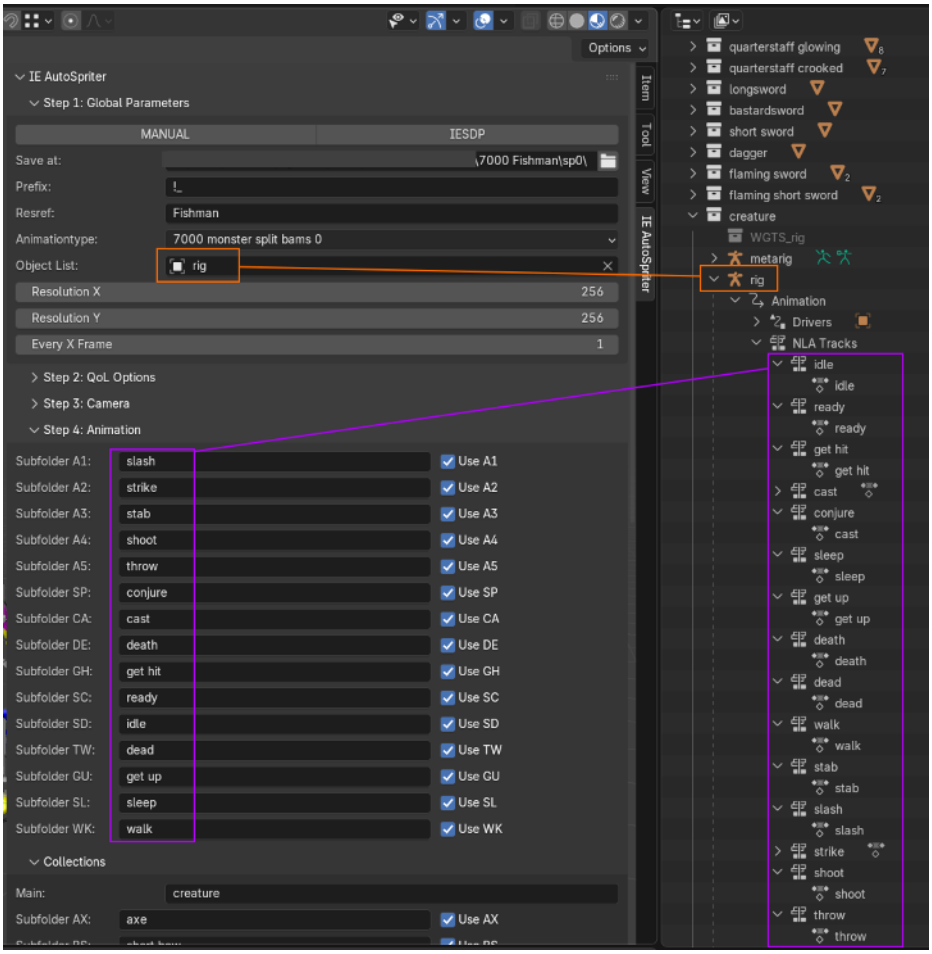
animation A4 is triggered.

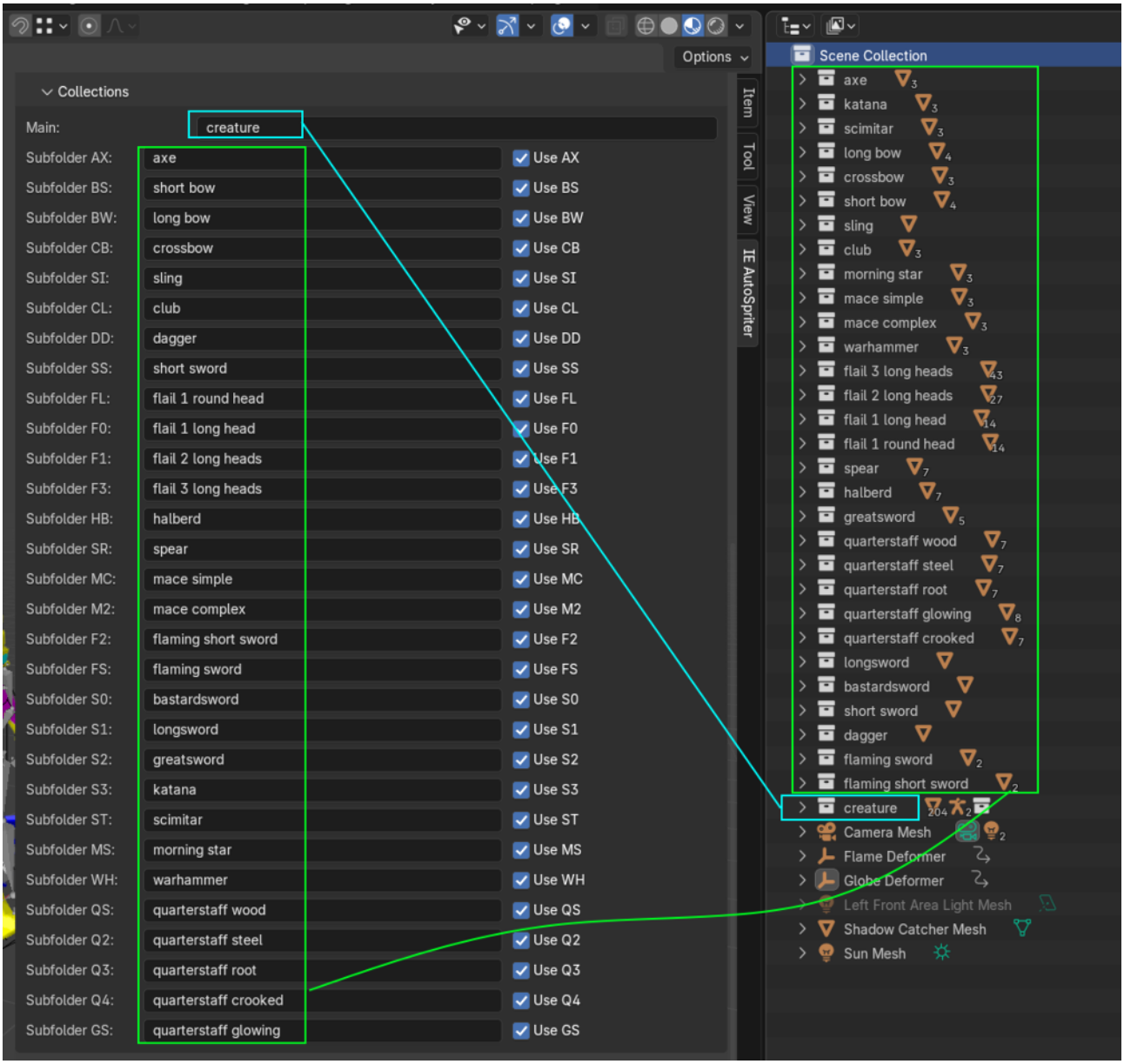
- G14: GH=27-35 (unused), DE=36-44, TW=45-53 has an unused TW.

The geometry node modifier is disabled for the viewport, but not for rendering, as it would otherwise consume a lot of processing power during animation. Be careful when enabling it.



Weapon Overlays



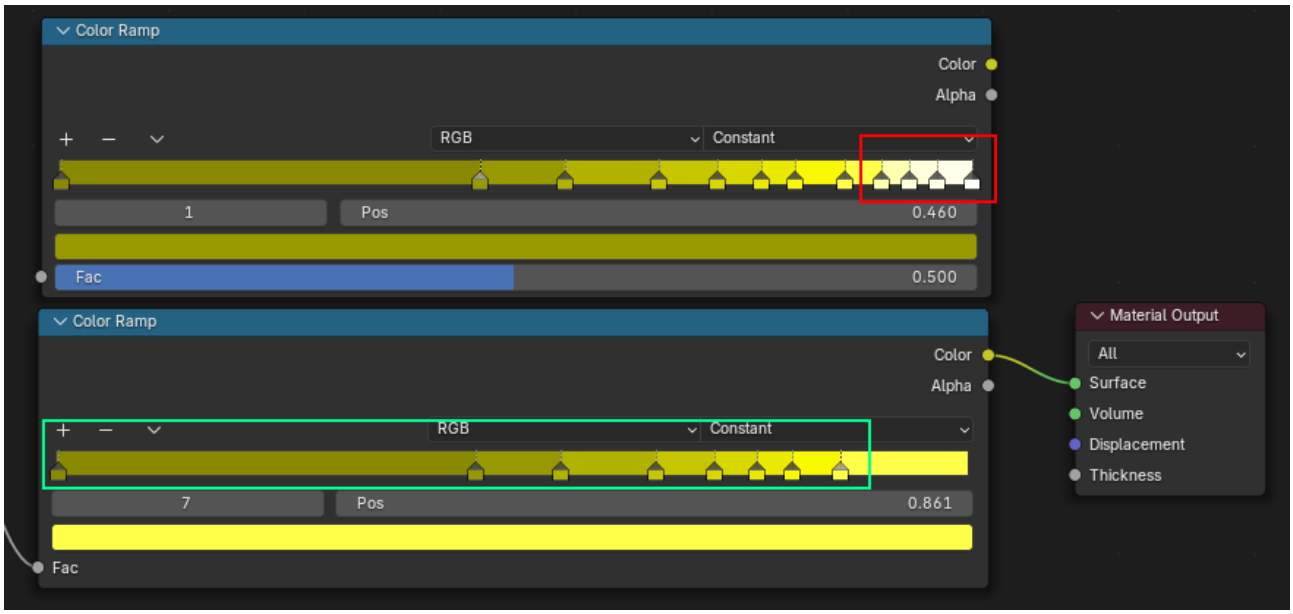


https://gibberlings3.github.io/iesdp/file_formats/ie_formats/itm_v1.htm

- SI = SL (Sling)
- SR = SP (Spear)
- ST = SC (Scimitar)

In IE AutoSpriter, these are renamed WOVLs ([resref][seq][wovl]?.BAM). This was done to prevent conflicts with standard animations like Ready (SC), Sleep (SL), and Conjure (SP).

Shading Nodes



In most textures, specifically regarding the color ramps added to restrict the palette to colors needed for IE color replacements, the very light colors have been removed. This was done for testing purposes and to minimize color bleeding(to prevent "bleed" into neighboring pixels).

Notes

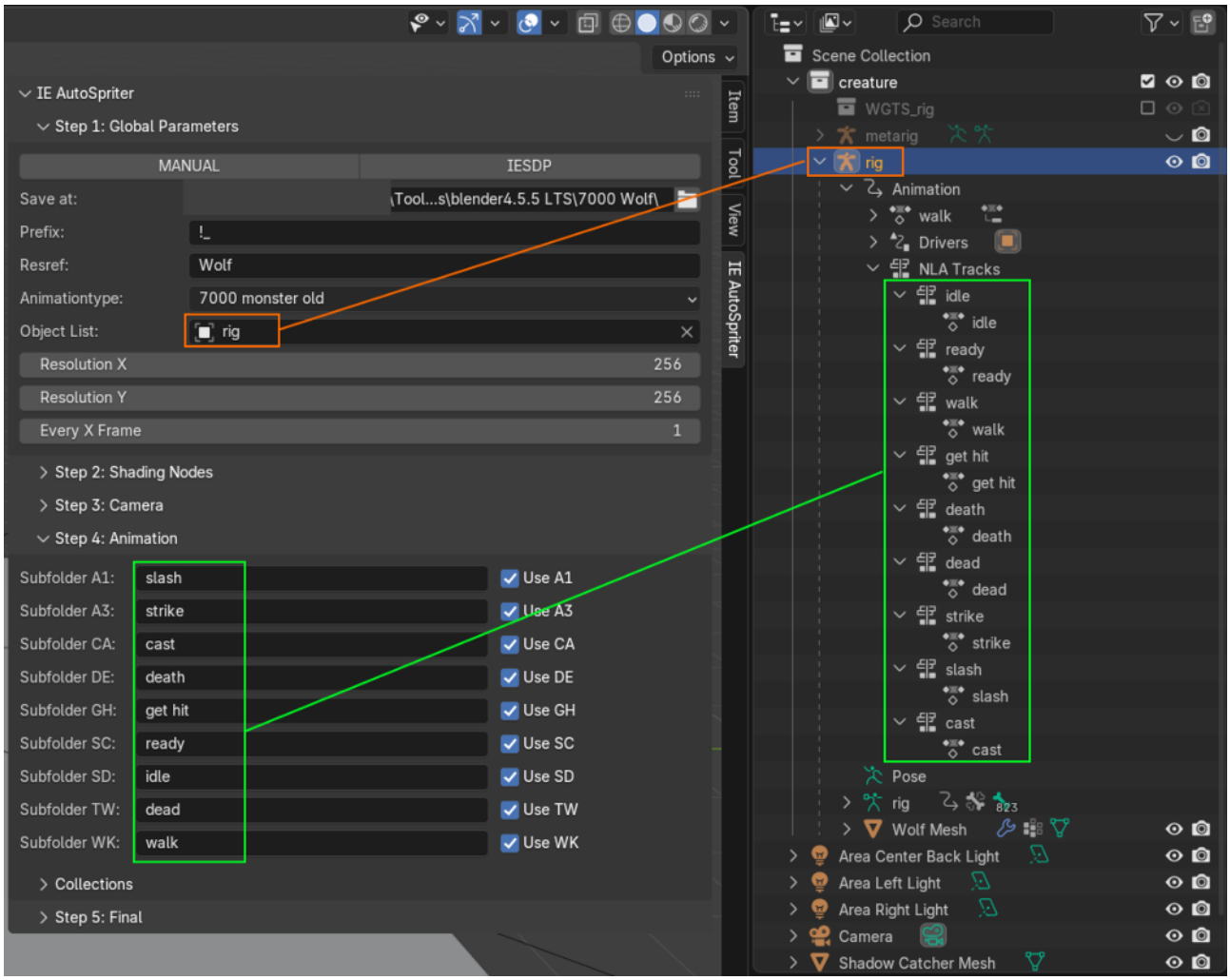
Another issue was that color bleeding (incorrectly positioned pixel colors on the model) occurred.

This issue was produced by the Near Infinity filter 'Resize BAM frames,' which can be found in the post-processing section.

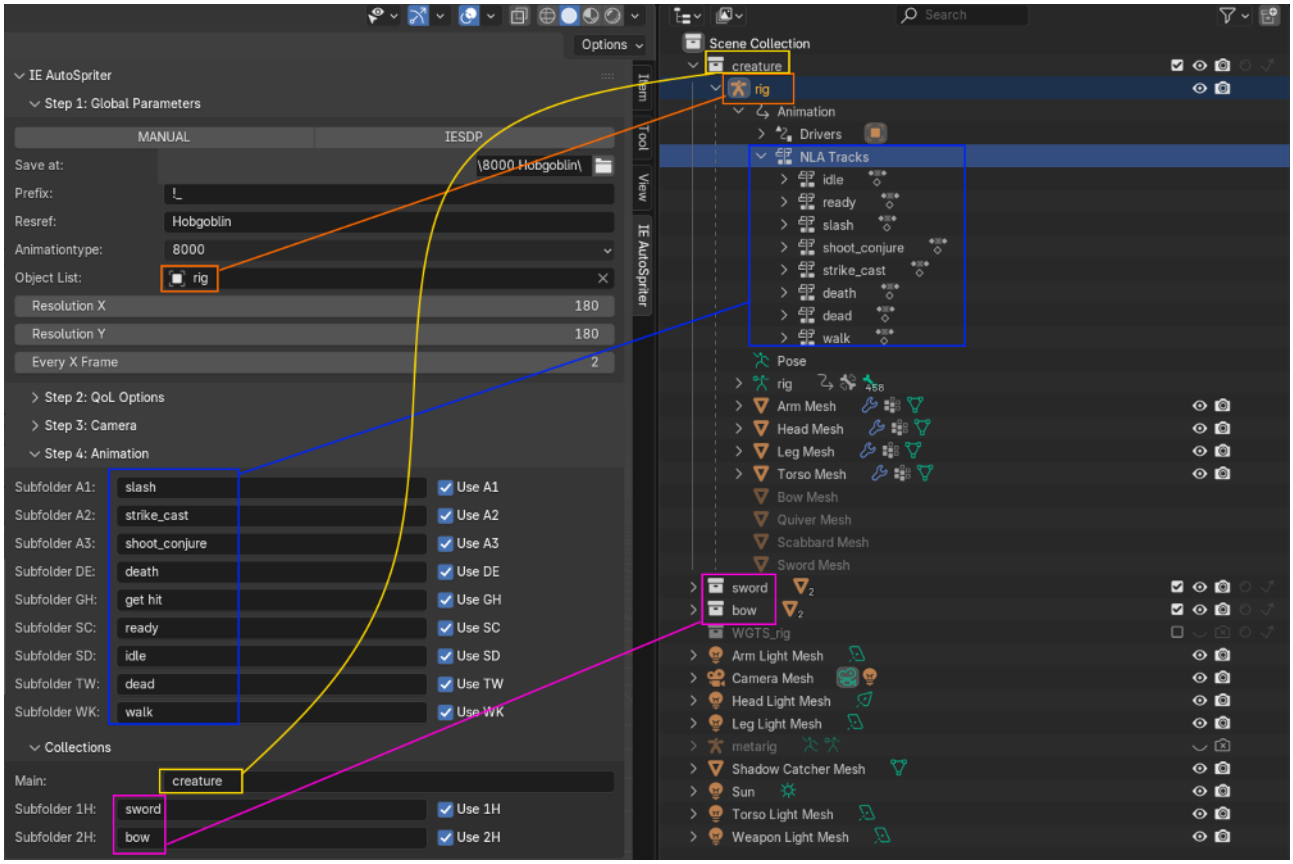
Solution:

- First, create the BAM file with the specific color palette (e.g., from LFAM.BAM for color replacements) WITHOUT any filters.
- Re-import the created BAM file(Replace) and then use the filter 'Resize BAM frames.' With that, the BAM file can be overwritten.

Animation Type 7000 Monster_old



Animation Type 8000(NEW)



Notes

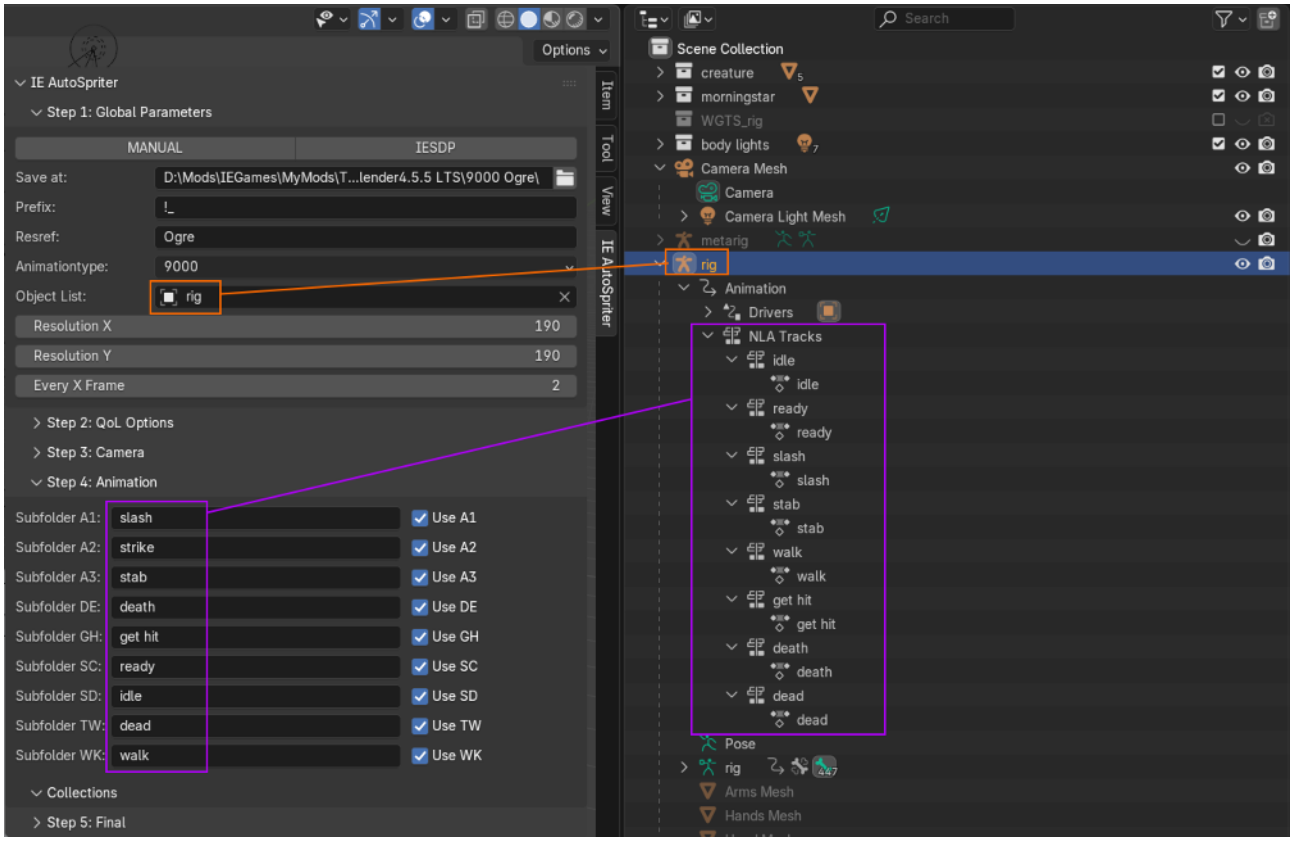
See animation Type C000 „Blender Settings“. Besides that the following setting is activated:

- Activate „Look“ to „Very High Contrast“(Properties window->Render->Color Management)

It seems, the weapons were layered differently in terms of their orientation for longbows it seems. Behaviour is triggered by the item "Left-handed" flag.

- 0 → S: WP=Front, CH=Back,
- 1 → SW: WP=Front, CH=Back,
- 2 → W: WP=Front, CH=Back,
- 3 → NW: WP=Front, CH=Back,
- 4 → N: WP=Back, CH=Front,
- 5 → NE: WP=Back, CH=Front,
- 6 → E: WP=Back, CH=Front,
- 7 → SE: WP=Back, CH=Front,

Animation Type 9000(NEW)



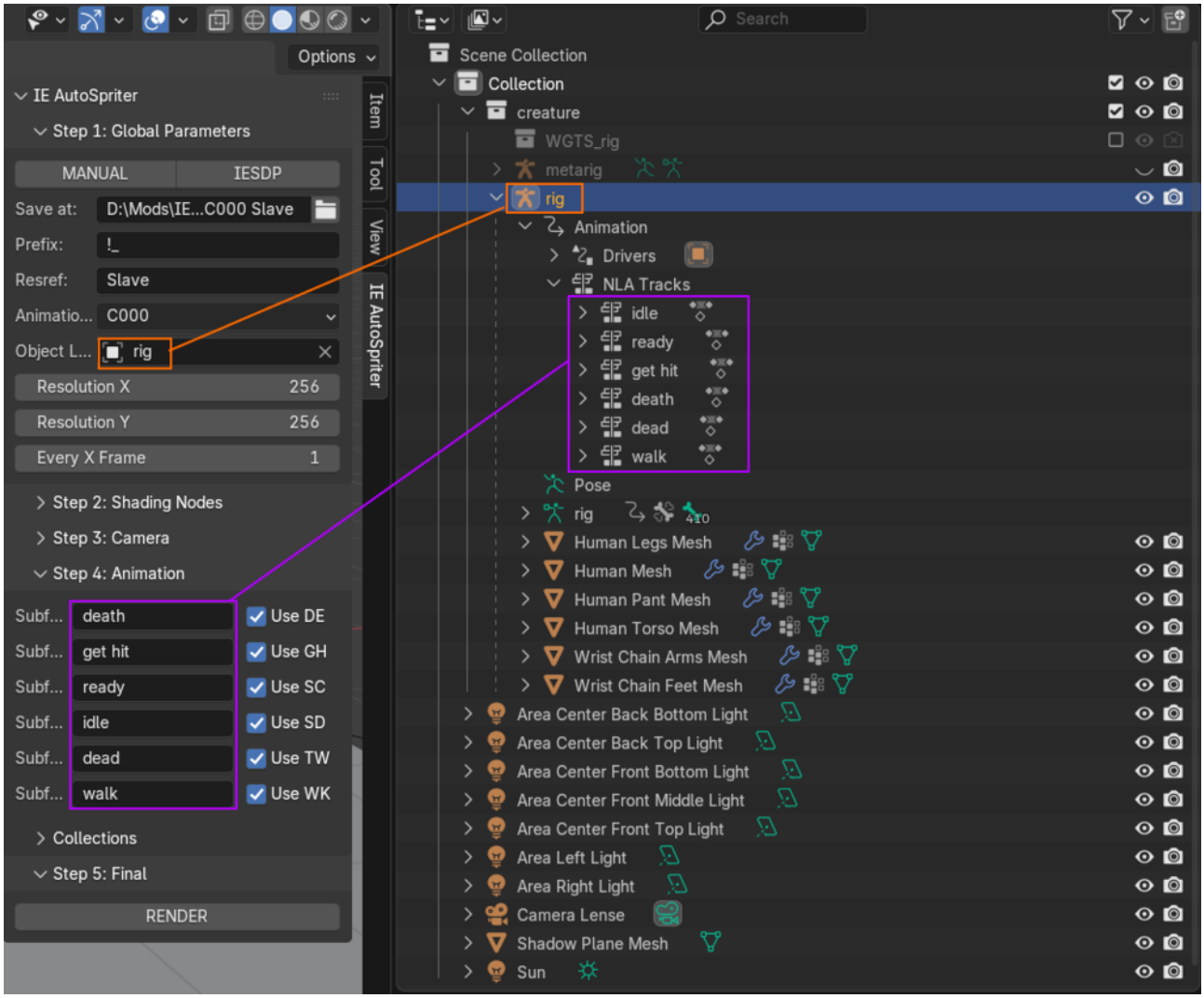
Similar blender ettings as in „Animation Type C000“ were used.

Notes

When the spell is being conjured, the SC(ready) animation plays. When the spell is cast, the A3 animation plays. This must be correctly noted in the IESDP.

Attack animations are not played for ranged weapons.

Animation Type C000



NOTE: Due to the new bug fix, some animations may be displayed incorrectly. The fix itself should work perfectly, however; this is due to a sloppy workflow during the demo's creation.

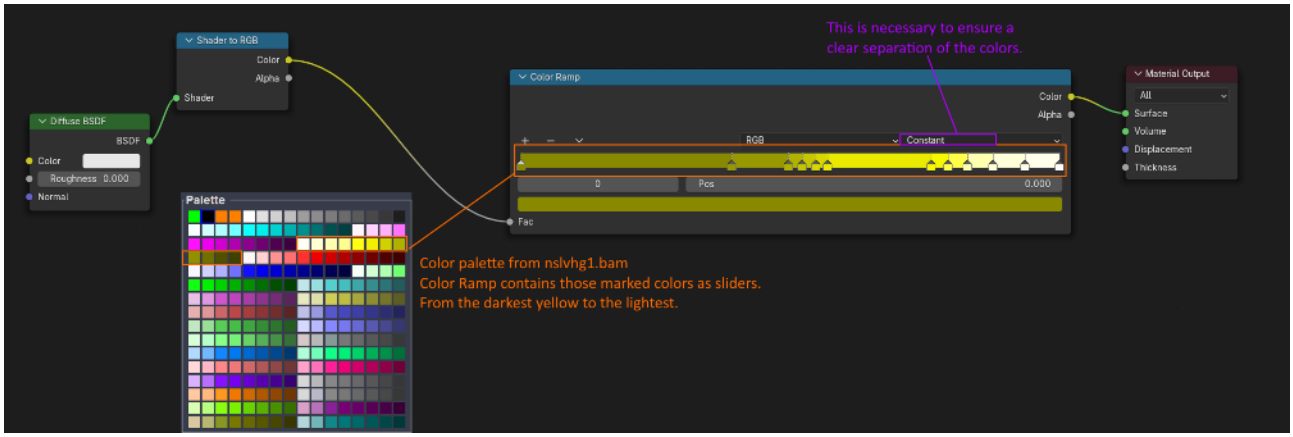
Blender Settings

- EEVEE is activated
- Deactivated Glossy for every light source(Properties window → Object → Visibility)
- Deactivated Dithering by setting it to 0(Properties window->Output->Post Processing)
- Changed the view Transform to Standard(Properties window->Render->Color Management)
- Deactivated Filter Size by setting it to 0px(Properties window → Render → Film)
- Changed the samplerate to 1 for the viewport(Properties window → Render → Sampling → Viewport)
- Changed the samplerate to 1 for the render(Properties

window → Render → Sampling → Render)

- Changed the world surface to black(Properties window → world → Surface → color)

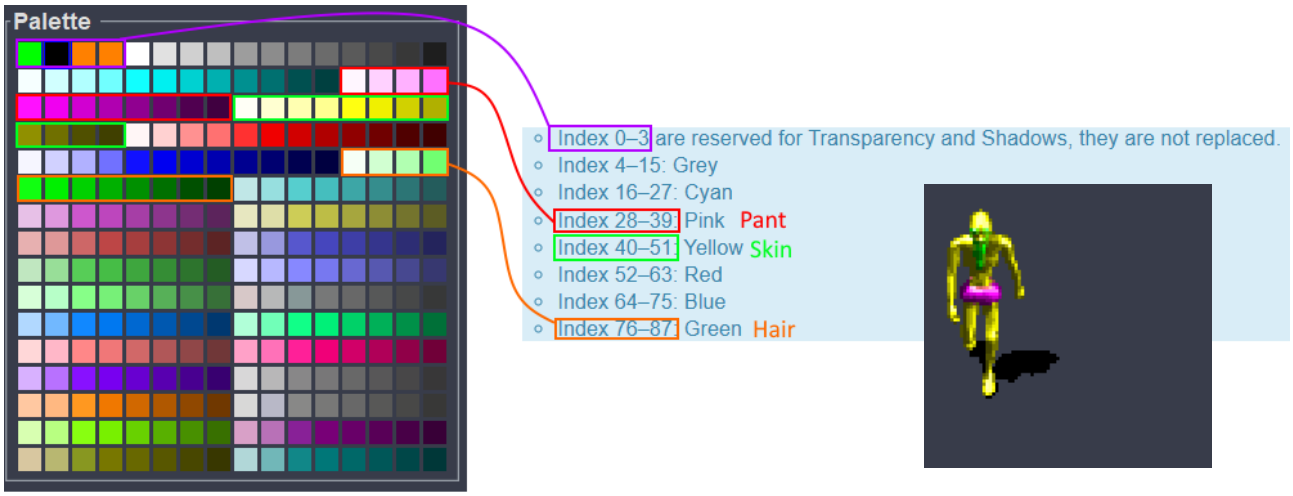
Shading Nodes



The image shows nodes representing skin tone. The same process was carried out for hair and pant color.

To optimally display the model's details, correct lighting is crucial, which in itself presents a challenge. Therefore, the model was divided into separate meshes, and some lights were linked to these meshes so that the lighting effect only affects those meshes.

Notes



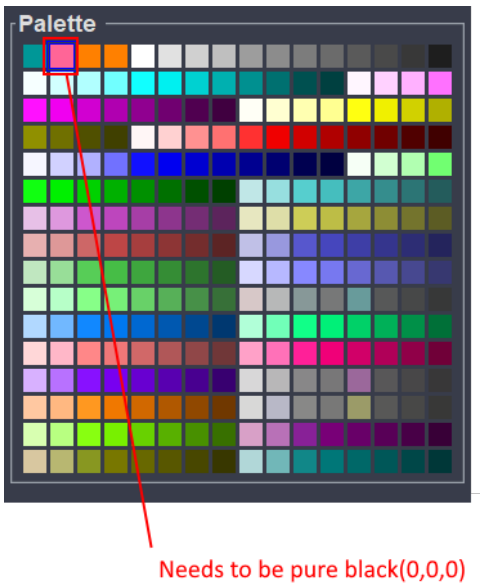
0 – 6: Avatar/Armor
- 00 → Belt Buckle (on armour) or amulets (on robes) – entire body for some animations 0=Grey
- 01 → Minor colour 1=Cyan
- 02 → Major colour 2=Magenta
- 03 → Skin colour 3=Yellow
- 04 → Strap colour (on armour) or minor trimming (on robes – the centre strip down a robe and the hem on some robes) 4=Red
- 05 → Armor colour (and major trimming on robe – cross trim and cape trim on Robe of Archmagi) 5=Blue
- 06 → Hair colour 6=Green

Source: https://gibberlings3.github.io/iesdp/opcodes/bgee.htm

Search for: C*yan/Teal, Magenta/Pink,*

„**nslvhg1.bam**“ color palette was used to create the demo bam test files.

Exporting the Color Palette from „**CDFB1A1.bam**“ and „**CIFT4SA.BAM**“ for the Type C000 demo model when converting into BAM with Near Infinity worked as well. It is very likely that almost any color palette from the game that contains modifiable color ranges will work. The necessary adjustments could be minor.

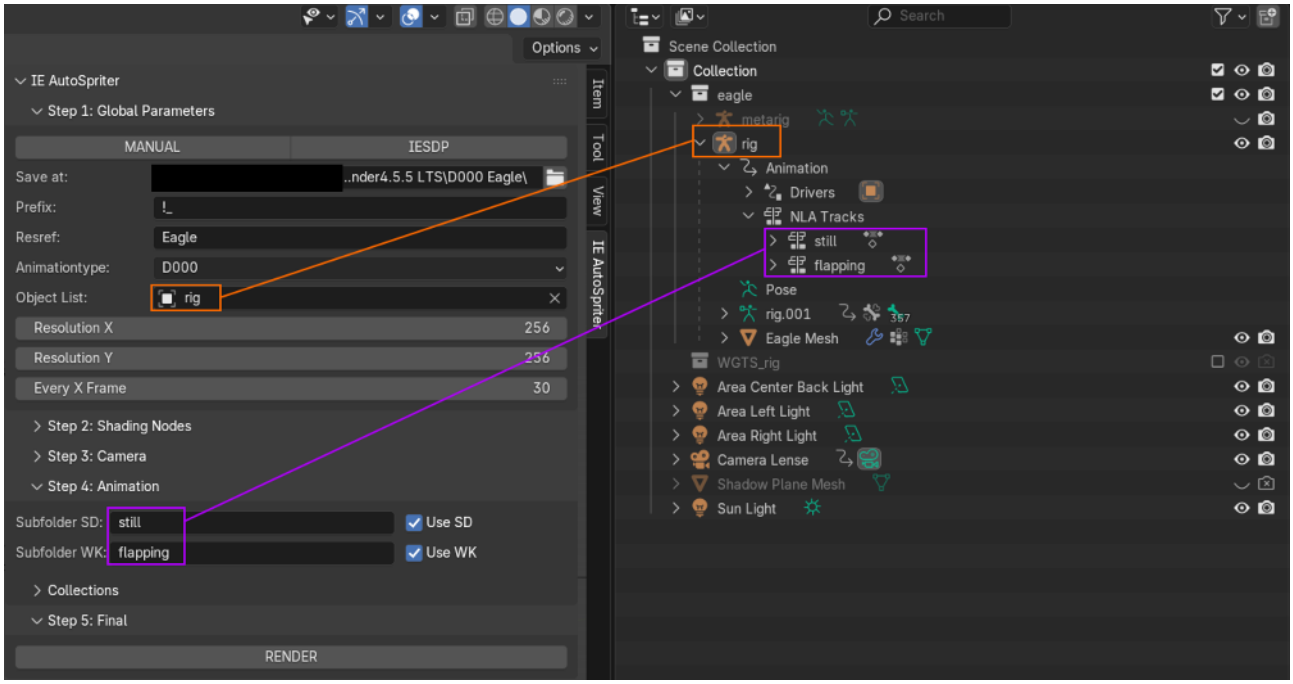


When loading the color palette from these BAM files, it is important that the color for index 1 is black so that the black shadow of the model is displayed transparently(If not then change it to balck).

Please note that these changes are NOT required for every sprite. The color palette is used for every sprite.

Also, using filters in the NI BAM Converter that change the colors of the model may also change the color palette of the saved BAM file.

Animation Type D000



For further information see „Animation Type B000“.

Troubleshooting/FAQ

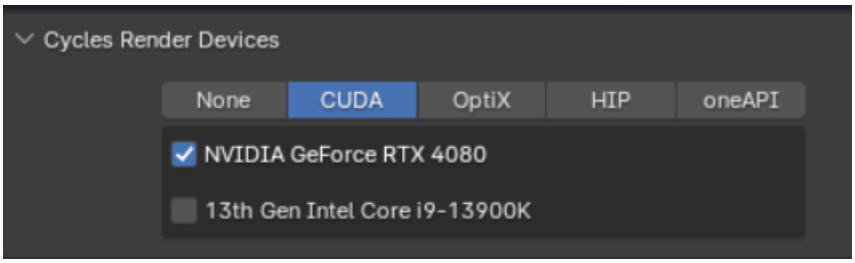
„Save at“ path empty, what happens?

This saves the animation folders and their contents directly to the Blender project path. However, it's more human-readable to include this path in the "**Save at** option.

How can time performance be increased?

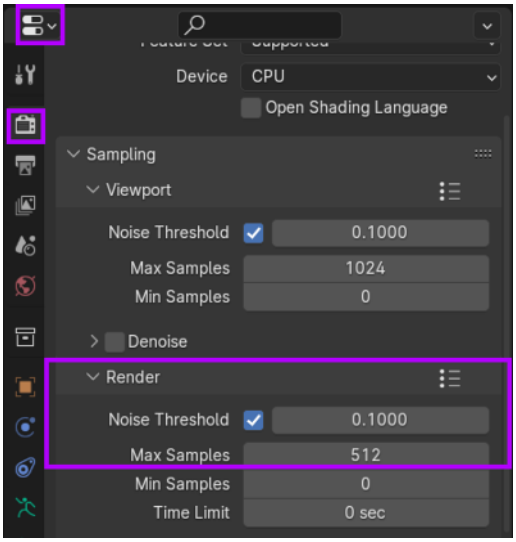
Testresults can be found here.

Usage of GPU power



This option can be found under Edit → Preferences... → System. However, the change didn't result in any performance improvements on my PC.

Alter sampling parameters



You can find this in the editor window under Render (camera icon) → Sampling → Render.

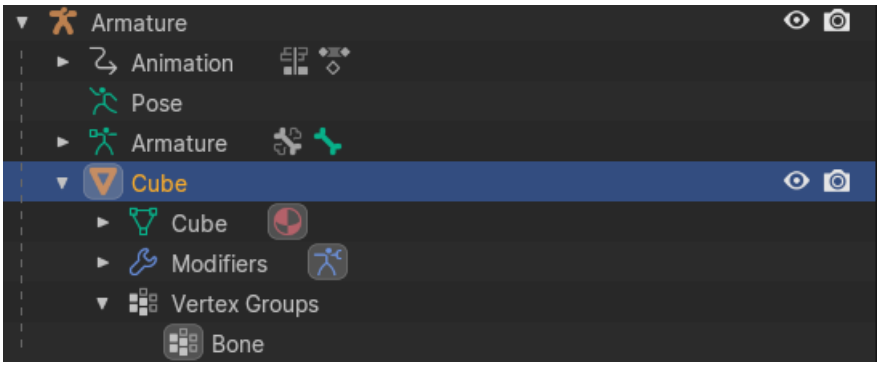
Increasing the "**Noise Threshold**" and reducing the "**Max. Samples**" should improve the time performance.

Further reducing the "**Max. Samples**" (e.g., to 256) should certainly be possible without any significant loss in sprite quality, but increasing the "**Noise Threshold**" may not be as effective. Users will have to test for themselves what works best for them.

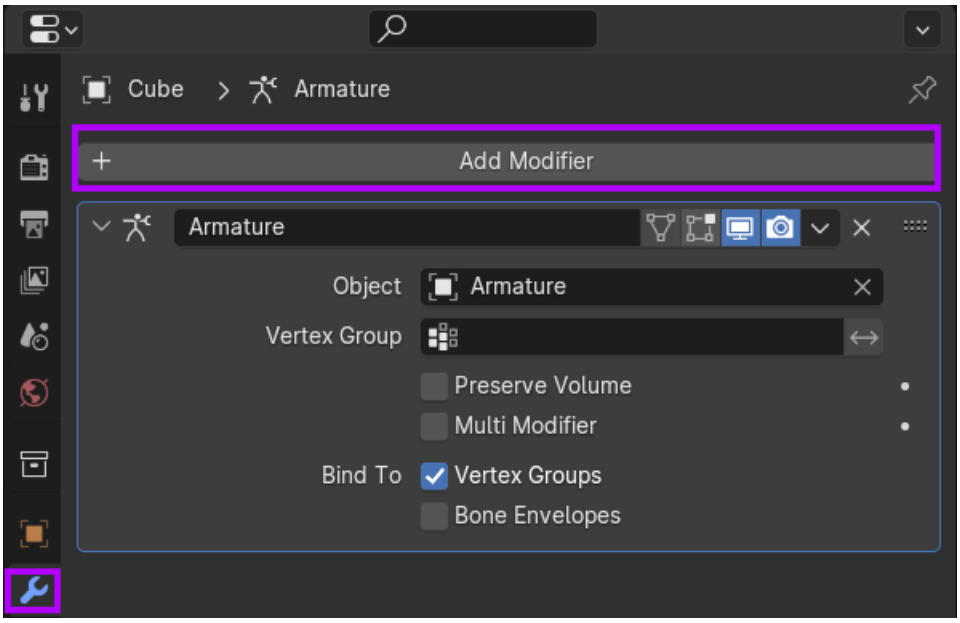
Use Decimate Modifier

This should only be considered as a last resort as it will definitely reduce model quality. Decimate Modifier can be used as follows:

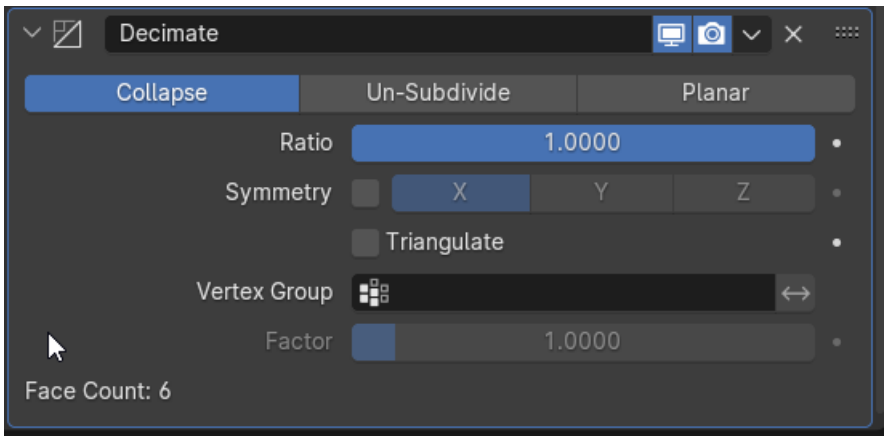
1. Choose your model in this particular case cube



2. Press „**Add Modifier**“ and select Generate → Decimate

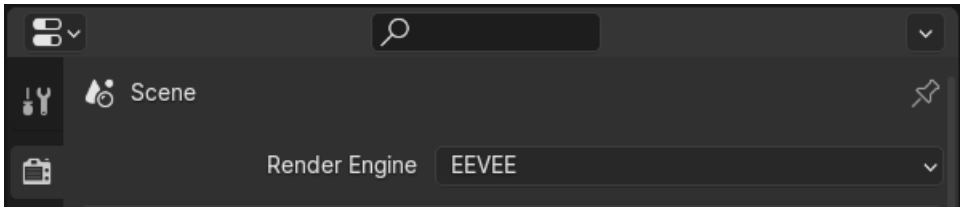


3. **Reducing Ratio** will improve performance time(see testexample)



Change Render Engine

Switching to EEVEE will significantly reduce rendering time~~, but this is not currently supported by the template save files that ship with IE AutoSpriter and may be added later in the development process.~~



Here is the specific IE AutoSpriter test result for EEVEE (v0.36.3-legacy-for-blender4.0)

Specific animation types spam new images in Image Editor

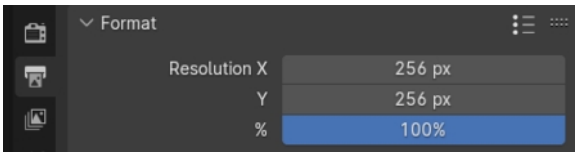
Yes, this is a known problem that should not have any side effects and can be solved as follows

- File → Clean Up → Recursive Unused Data-Blocks

Note this will also delete things that are not used but maybe created for later use, unless „Fake User“ is active for that specific object.

How does the resolution setting work?





If the resolution setting in IE AutoSpriter is clicked to be manipulated then it will have its values transferred to blender's render settings in “Output Properties”. If the values are different in both setting(IE AutoSpriter and “Output Properties”) then the one from “Output Properties” will be transferred to the resolution settings in IE AutoSpriter after “RENDER” button in step 5 is pressed. This is to prevent having the IE AutoSpriter resolution setting constantly manipulating the “Output Properties” resolution if the user prefers to use the resolution of “Output Properties” (e.g. for camera view).

Don’t use special characters for the string properties



Since some string properties are used to create folders, a special character such as “/” will result in the creation of a subfolder.

Blender terminated unexpectedly during rendering

Forcibly exiting Blender while rendering is unfortunate and may cause issues in the settings of the collections and meshes IE AutoSpriter needs and manipulates.

While testing, such an issue occurred only once; however, it is worth checking the holdout button for the collection. Deactivating it within the collections AutoSpriter needs to use can help. Anything else falls under general troubleshooting.

ALWAYS make a backup file of your project before rendering with IE AutoSpriter.